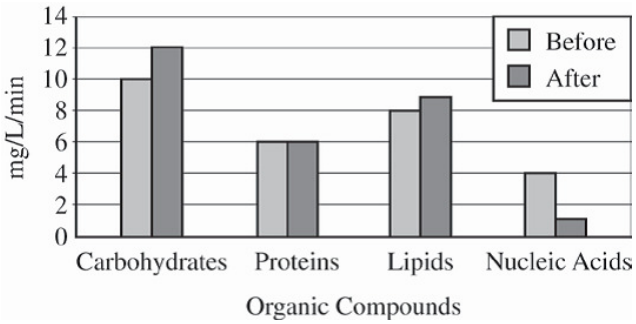
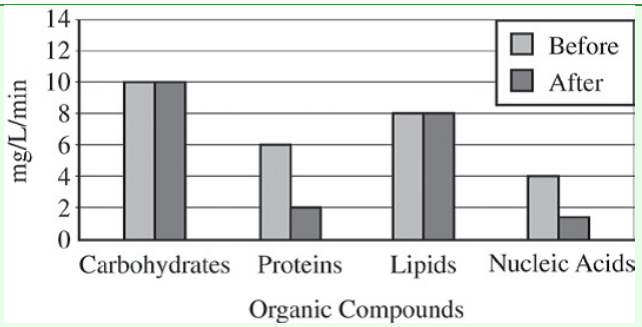
1 Chemistry of Life MCQ

1. A culture of *Spirogyra* (an autotrophic alga) is maintained in a water solution containing dissolved carbon dioxide and a source of phosphates but lacking nitrogen compounds. A researcher determines the rates of synthesis of several organic compounds found in the *Spirogyra* before and after several weeks in the water solution. Which of the following graphs best illustrates a likely result of the experiment?

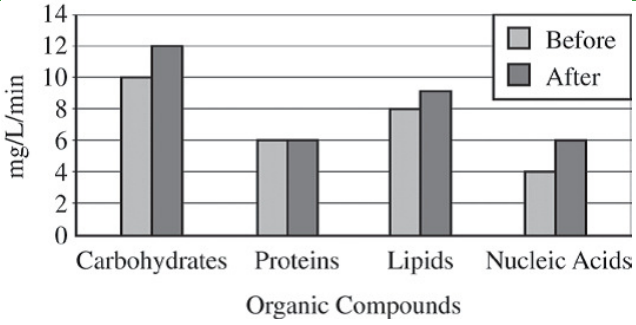
(A)



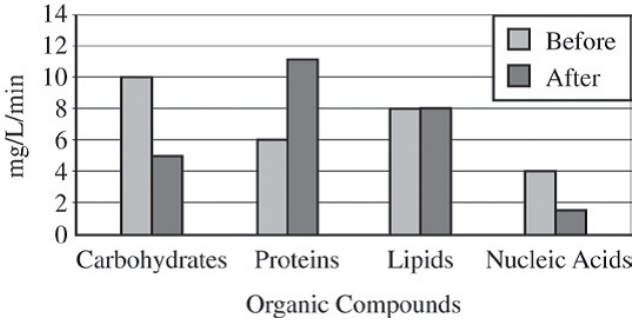
(B) ✓



(C)



(D)



1 Chemistry of Life MCQ

Answer B

This option is correct. Without a source of nitrogen, the *Spirogyra* cannot synthesize proteins or nucleic acids, both of which contain nitrogen. This answer reflects that the student is able to represent graphically the exchange of molecules between an organism and its environment and the use of these molecules in synthesis.

2. A feature of organic compounds NOT found in inorganic compounds is the presence of

(A) ionizing chemical groups

(B) electrons

(C) carbon atoms covalently bonded to each other ✓

(D) oxygen

(E) hydrogen bonds

3. tRNA molecules deliver amino acids to ribosomes where the amino acid next

(A) forms an ionic bond with another amino acid

(B) is deaminated and catabolized

(C) becomes an anticodon

(D) is covalently bonded to a growing polypeptide ✓

(E) forms hydrogen bonds with esters

4. The CFTR protein is made up of 1,480 amino acids linked together in a chain. Some humans produce a version of the CFTR protein in which phenylalanine (an amino acid) has been deleted from position 508 of the amino acid chain.

Which of the following best predicts how the amino acid deletion will affect the structure of the CFTR protein?

(A) It will have no observable effect on the structure of the CFTR protein.

(B) It will affect the primary structure of the CFTR protein, but the other levels of protein structure will not be affected.

(C) It will affect the secondary and tertiary structures of the CFTR protein, but the primary structure will not be affected.

(D) It will affect the primary, secondary, and tertiary structures of the CFTR protein. ✓

Answer D

Correct. A change in the primary structure of the CFTR protein will lead to changes in the secondary

1 Chemistry of Life MCQ

and tertiary structures. As a result, the protein will not fold properly and will not function normally.

5. Which of the following describes a key difference among the 20 amino acids that are used to make proteins?

(A) Only some amino acids have an R-group.

(B) Only some amino acids have a carboxyl group (COOH).

(C) Some amino acids are hydrophobic. ✓

(D) Some amino acids contain the element phosphorus.

Answer C

Correct. Due to having nonpolar R-groups, 10 of the 20 amino acids are hydrophobic. Interactions between hydrophobic amino acids play an important role in determining protein structure and function.

6. Scientists examined the folded structure of a purified protein resuspended in water and found that amino acids with nonpolar R groups were primarily buried in the middle of the protein, whereas amino acids with polar R groups were primarily on the surface of the protein. Which of the following best explains the location of the amino acids in the folded protein?

(A) Polar R groups on the surface of the protein can form ionic bonds with the charged ends of the water molecules.

(B) Polar R groups are too bulky to fit in the middle of the protein and are pushed toward the protein's surface.

(C) Nonpolar R groups that cannot form hydrogen bonds with water are pushed into the middle of the protein. ✓

(D) Nonpolar R groups from different parts of the protein form covalent bonds with each other to maintain the protein's structure.

1 Chemistry of Life MCQ

7. Figure 1 shows three amino acids that are part of a polypeptide chain. Figure 2 shows the same section of the chain after a mutation has occurred.

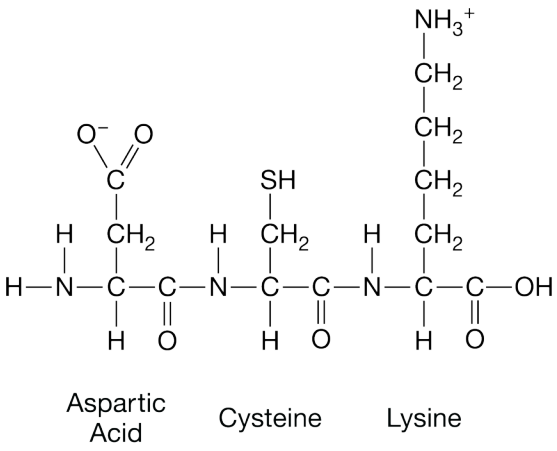


Figure 1: Original Amino Acid Chain

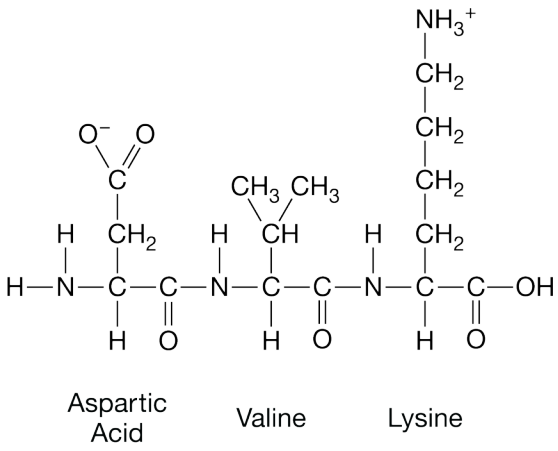


Figure 2: Mutated Amino Acid Chain

How might this change affect the structure and function of the protein?

(A) The R-group of the new amino acid, valine, has different chemical properties than the R-group of cysteine. This will cause the protein to misfold and not function properly in the cell. ✓

(B) The new amino acid, valine, has replaced cysteine in the new protein. Since the number of amino acids has remained the same, there will be no change in the three-dimensional folding, or function, of the protein.

(C) Since this is a linear section, it does not influence protein folding. Thus, there will be no change in protein structure or function.

(D) Since the new amino acid is bounded on one side by an amino acid with a negatively charged R-group and by an amino acid on the other side with a positively charged R-group, the charges will balance and the protein will fold as usual.

Answer A

Correct. Three-dimensional folding of a protein is due to interactions among the R groups of the amino acids. Cysteine has a sulfhydryl group, which may form a disulfide bridge with another part of the polypeptide chain. Valine has no sulfhydryl group and is nonpolar, which will affect how the polypeptide will fold.

1 Chemistry of Life MCQ

Students investigated the effect of light on the carbon cycle in aquatic ecosystems by performing the controlled experiment summarized below. The students placed equal amounts of water (pH 7.0) from a large aquarium in glass beakers. The students transferred aquatic plants from the aquarium to several of the beakers, and then they placed equal numbers of the beakers in the light or the dark (Figure 1: groups I and II). Similarly, the students transferred goldfish from the same aquarium to other beakers, and then they placed equal numbers of those beakers in the light or dark (Figure 1: groups III and IV). Finally, the students placed an equal number of beakers containing water only in the light or dark (Figure 1: groups V and VI).

After exposing the samples to light or dark for one hour, the students recorded the pH of the water in each beaker. Carbon dioxide dissolved in water will lower the pH of an aqueous solution. In the experiment, the students used changes in pH to monitor changes in the amount of carbon dioxide in the water. For each treatment group, the students calculated the mean pH and standard error, as documented in the table below.

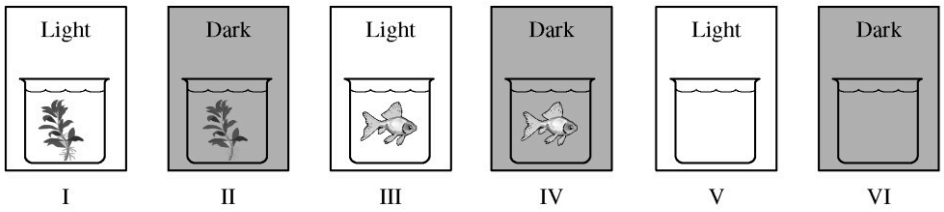


Figure 1. Treatment groups

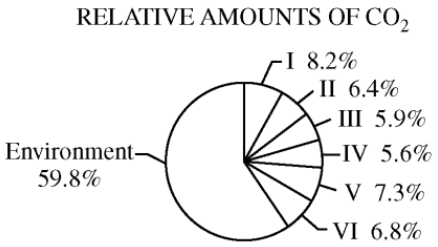
MEAN pH OF TREATMENT GROUPS AFTER 1 HOUR

Treatment group (n = 10)	I	II	III	IV	V	VI
Mean pH	8.2	6.4	5.9	5.6	7.3	6.8
Standard error of the mean	0.3	0.3	0.2	0.3	0.5	0.4

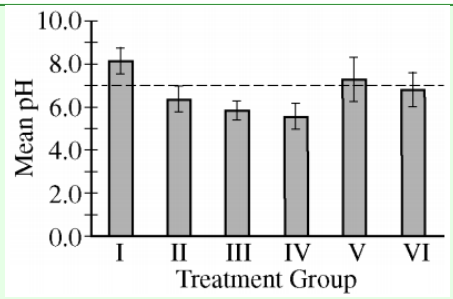
8. Which of the following graphs is the most appropriate representation of the experimental results documented in the table?

1 Chemistry of Life MCQ

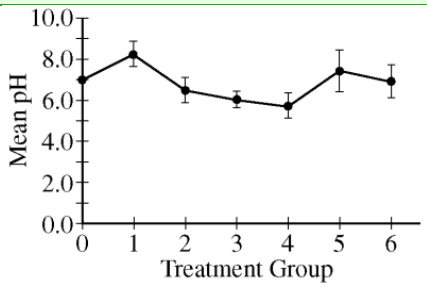
(A)



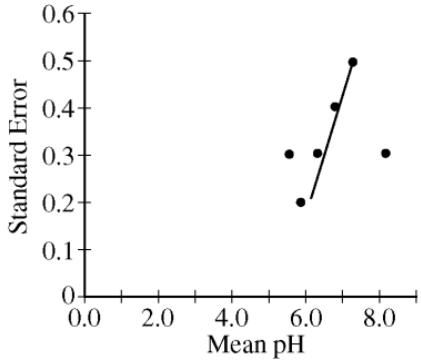
(B) ✓



(C)



(D)



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9. Bacteriophages are viruses that infect bacteria. In an experiment, bacteriophages were labeled with either radioactive phosphorus or radioactive sulfur. The labeled bacteriophages were incubated with bacteria for a brief amount of time and then removed. The infected bacteria cells were found to contain significant amounts of radioactive phosphorus but not radioactive sulfur.

Based on the results of the experiment, which of the following types of molecules did the bacteriophages most likely inject into the bacteria cells?

1 Chemistry of Life MCQ

(A) Simple carbohydrate

(B) Amino acid

(C) DNA ✓

(D) Polypeptide

10.

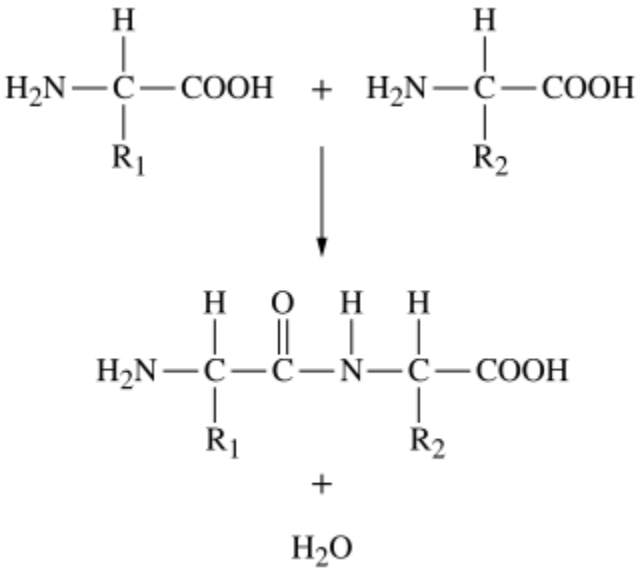


Figure 1. A representation of a reaction involved in the formation of a biological molecule

Which of the following best explains the reaction represented in Figure 1 ?

(A) It is a dehydration synthesis reaction involved in the formation of a lipid.

(B) The reaction is the hydrolysis of a nucleic acid and results in the formation of a nucleotide.

(C) The reaction is the hydrolysis of a nucleic acid and results in the formation of a phospholipid.

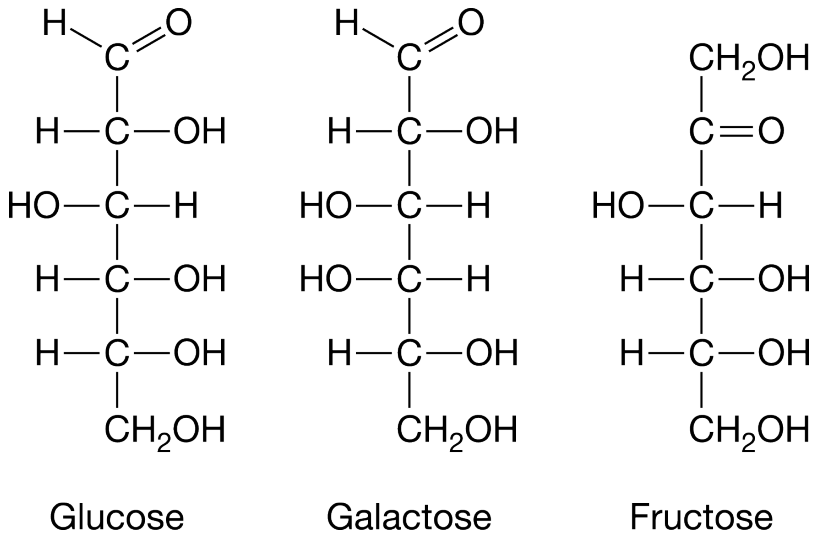
(D) It is a dehydration synthesis reaction involved in the formation of a polypeptide. ✓

Answer D

Correct. Two amino acids undergo a dehydration synthesis reaction, resulting in the formation of a peptide bond between them, the process involved in the growth of a polypeptide.

1 Chemistry of Life MCQ

11. The carbohydrates glucose, galactose, and fructose have the same chemical formula $(C_6H_{12}O_6)$ but different structural formulas, as represented in the figure.



Which of the following statements about glucose, galactose, and fructose is most likely true?

(A) The carbohydrates have the same properties because they have the same number of carbon, hydrogen, and oxygen atoms.

(B) The carbohydrates have the same properties because they each have a single carbon-oxygen double bond.

(C) The carbohydrates have different properties because they have different arrangements of carbon, hydrogen, and oxygen atoms. ✓

(D) The carbohydrates have different properties because they have different numbers of carbon-carbon bonds.

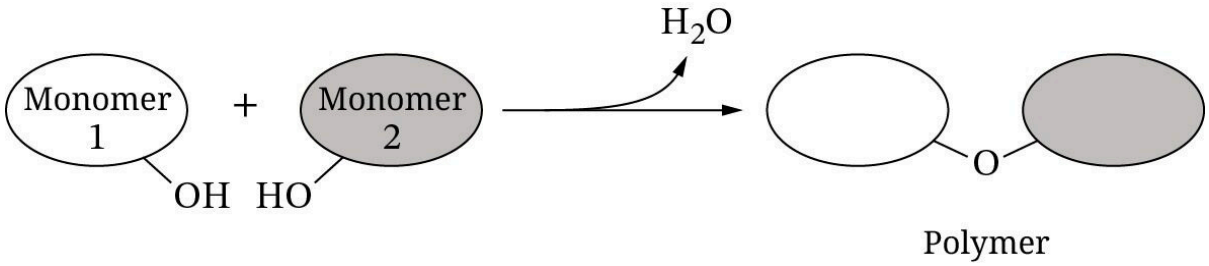
Answer C

Correct. Compounds that have the same chemical formula but different structural formulas usually have different properties.

1 Chemistry of Life MCQ

12. Figure 1 shows the formation of a carbohydrate polymer.

Figure 1. Formation of a carbohydrate polymer



Which of the following best describes how the carbohydrate polymer is being built in Figure 1?

(A) A water molecule is added to form a hydrogen bond between two monomers.

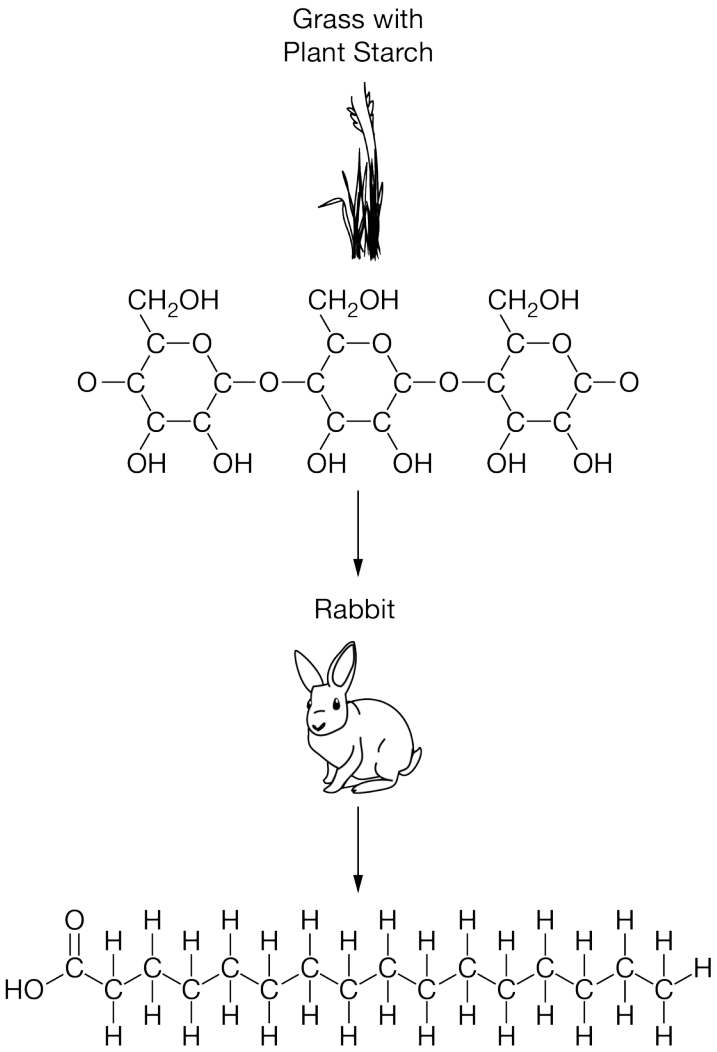
(B) A water molecule is added to form a covalent bond between two monomers

(C) A water molecule is removed to form a hydrogen bond between two monomers.

(D) A water molecule is removed to form a covalent bond between two monomers. ✓

1 Chemistry of Life MCQ

13.



Which of the following statements best describes how organisms such as rabbits obtain the carbon necessary for building biological molecules?

(A) Rabbits eat plants and use energy absorbed from the plants to make carbon atoms from electrons, protons, and neutrons in the air.

(B) Rabbits eat plants and break down plant molecules to obtain carbon and other atoms that they rearrange into new carbon-containing molecules. ✓

(C) Rabbits eat plants and use water absorbed from the plants to hydrolyze CO_2, which the rabbits breathe in from the air and use as a carbon source.

(D) Rabbits eat plants and make carbon-containing molecules by using carbon atoms that the plants absorbed from the soil and stored in the cells of their leaves.

1 Chemistry of Life MCQ

Answer B

Correct. Rabbits break down the plant molecules to obtain the individual atoms or small molecules that can be built into larger, complex biological molecules.

Different polysaccharides are used by plants for energy storage and structural support. The molecular structures for two common polysaccharides are shown in Figure 1. Starch is used by plants for energy storage, and cellulose provides structural support for cell walls. The monomer used to construct both molecules is glucose.

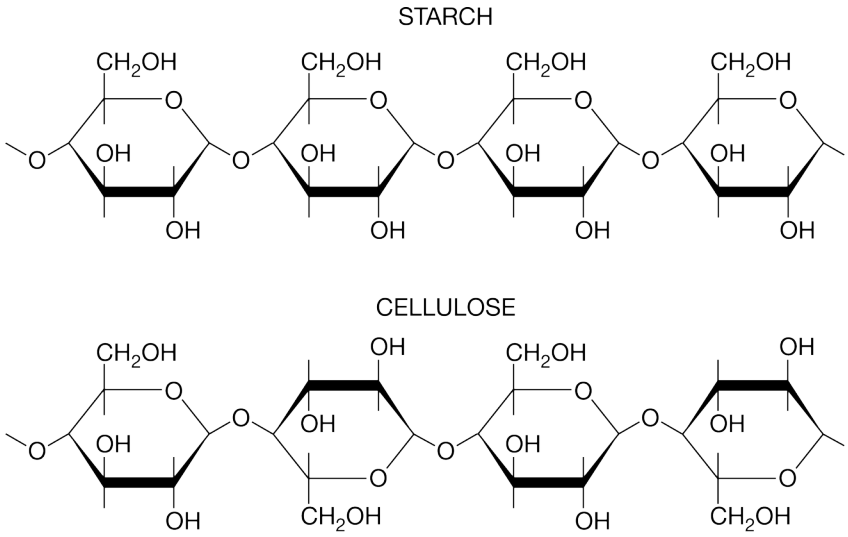


Figure 1. Comparison of segments of starch and cellulose

A study determined the effect of two different digestive enzymes, A and B, on these two polysaccharides. Table 1 presents the data from the study.

Table 1. Effect of Enzymes A and B on Cellulose and Starch

Test Tube	Polysaccharide Added	Enzyme Added	Glucose Detected after 5 Minutes at 37°C
1	Cellulose	A	No
2	Cellulose	B	Yes
3	Starch	A	Yes
4	Starch	B	No

Mammals do not produce digestive enzyme B. However, sheep and cattle are two types of mammals that contain microorganisms in their digestive tract that produce enzyme B.

1 Chemistry of Life MCQ

14. Which of the following would most likely occur if cattle lost the ability to maintain a colony of microorganisms in their digestive tract?

(A) Cattle would no longer be able to synthesize cellulose.

(B) Cattle would have to convert cellulose to starch before digesting it.

(C) Cattle would have to start producing enzyme B without the help of the bacteria.

(D) Cattle would no longer be able to use cellulose as a primary source of glucose. ✓

Answer D

Correct. Without the enzyme B produced by microorganisms in their digestive tract, cellulose would pass through the digestive tract without being digested.

15. Which of the following best describes the process that adds a monosaccharide to an existing polysaccharide?

(A) The monosaccharide is completely broken down by a specific enzyme and then the atoms are reorganized and made into a polysaccharide.

(B) Ionic bonds are formed between adjacent carbon atoms of the monosaccharide and the polysaccharide by adding water $(\mathrm{H_2O})$ and a specific enzyme.

(C) A specific enzyme removes the hydrogen (H) from the monosaccharide and the hydroxide (OH) from the polysaccharide, creating a bond between the two and creating a water $(\mathrm{H_2O})$ molecule. ✓

(D) A specific enzyme removes two hydroxides (OH), one from the monosaccharide, and one from the polysaccharide, creating a bond between the two monosaccharides and creating a hydrogen peroxide ($\mathrm{H_2O_2}$) molecule.

Answer C

Correct. This is a description of dehydration synthesis, which joins multiple monosaccharides to create a polysaccharide and produces water $(\mathrm{H_2O})$ molecules.

16. Based on Figure 1, which of the following best compares the atomic structures of starch and cellulose?

(A) Starch is composed of carbon, hydrogen, and oxygen, while cellulose also contains nitrogen.

(B) Starch and cellulose are composed of repeating glucose monomers; however, in cellulose every other glucose monomer is rotated 180 degrees. ✓

(C) Starch is composed of monomers that each have a $\mathrm{CH_2OH}$ group, while cellulose only has a $\mathrm{CH_2OH}$ group on every other monomer.

(D) Starch and cellulose are composed of identical monomers and therefore have identical structures.

1 Chemistry of Life MCQ

Answer B

Correct. Both starch and cellulose are both composed of repeating glucose molecules; however, the orientation of every other glucose in cellulose is upside down compared with the ones on either side.

17. Which of the following statements best describes the different functions of starch and cellulose in plants?

(A) The differences in the assembly and organization of the monomers of these two polymers result in different chemical properties. ✓

(B) Since starch and cellulose are composed of identical monomers, the cellular environment where they are located controls their function.

(C) The monomers of cellulose are connected by covalent bonds, making it idea for structural support.

(D) The monomers of starch are connected by ionic bonds, making it ideal for energy storage for plants.

Answer A

Correct. The identical orientations of the glucose monomers in starch create a polysaccharide with alpha bonds that is easy to break down into glucose for energy use. The alternating orientations of the glucose monomers in cellulose create beta bonds that produce a rigid polymer that is difficult to digest for energy use.

---

18. Cholinesterase is an enzyme that catalyzes the hydrolysis of the signaling molecule acetylcholine. Which of the following best explains how cholinesterase carries out its function?

(A) Cholinesterase breaks covalent bonds within acetylcholine by the addition of a water molecule. ✓

(B) Cholinesterase breaks covalent bonds within acetylcholine by removing a water molecule.

(C) Cholinesterase forms covalent bonds between acetylcholine and another molecule by the addition of a water molecule.

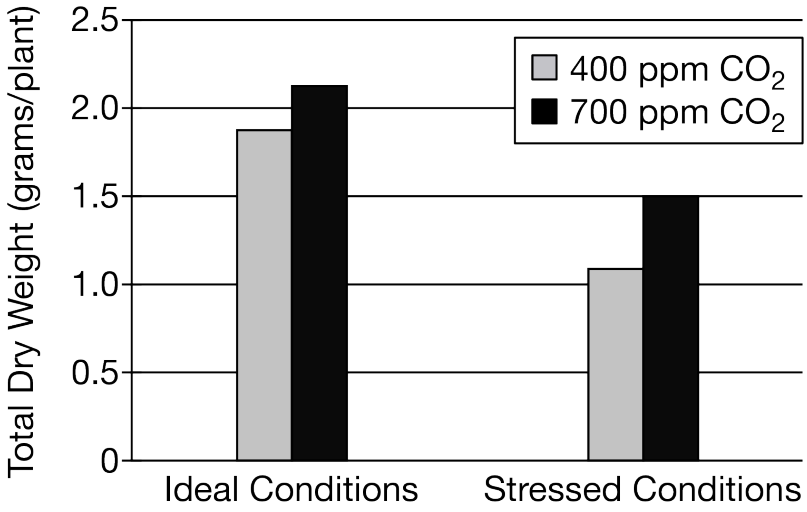
(D) Cholinesterase forms covalent bonds between acetylcholine and another molecule by removing a water molecule.

Answer A

Correct. Hydrolase enzymes catalyze the cleavage of covalent bonds by adding a water molecule to the bond.

1 Chemistry of Life MCQ

19. The figure shows the results of an experiment to investigate the effects of an enriched CO_2 environment on plant growth. Identical plants were separated into different groups and grown either in a standard CO_2 environment (400 ppm CO_2) or in an enriched CO_2 environment (700 ppm CO_2). Of the plants in each environment, half were grown under ideal conditions and half were grown under stressed conditions.



Based on the figure, which statement best describes the observed relationship between atmospheric CO_2 enrichment and plant growth under ideal and stressed conditions?

(A) The increase in atmospheric CO_2 had no observable effect on plant growth under either ideal or stressed conditions.

(B) The increase in atmospheric CO_2 resulted in a greater increase in plant growth under ideal conditions than under stressed conditions.

(C) The increase in atmospheric CO_2 resulted in a greater increase in plant growth under stressed conditions than under ideal conditions. ✓

(D) The increase in atmospheric CO_2 resulted in an inhibition of plant growth under both ideal and stressed conditions.

Answer C

Correct. Based on the figure, the increase in atmospheric CO_2 resulted in a greater increase in plant growth under stressed conditions than under ideal conditions.

20. Which of the following is responsible for the cohesive property of water?

1 Chemistry of Life MCQ

(A) Hydrogen bonds between the oxygen atoms of two adjacent water molecules

(B) Covalent bonds between the hydrogen atoms of two adjacent water molecules

(C) Hydrogen bonds between the oxygen atom of one water molecule and a hydrogen atom of another water molecule ✓

(D) Covalent bonds between the oxygen atom of one water molecule and a hydrogen atom of another water molecule

(E) Hydrogen bonds between water molecules and other types of molecules

21. Which of the following communities is likely to be most stable?

(A) One where a keystone predator has been removed

(B) One with uniformly spaced vegetation

(C) One with high species diversity ✓

(D) One that lacks decomposers

22. Researchers compared similar proteins from related organisms in different habitats. They found that the proteins from organisms living in harsh environments had a greater number of cysteine amino acids than did proteins from organisms not living in harsh environments. The structure of cysteine is shown. Bonds can form between the sulfur atom of different cysteine amino acids (S-S bonds).

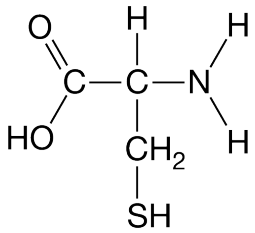


Figure 1. Chemical structure of cysteine

Which of the following best describes the effect of a greater number of cysteine amino acids on the stability of the proteins?

(A) The change has no effect on the stability of the protein because only one type of amino acid is involved.

(B) The change leads to increased protein stability because of an increased number of S-S bonds in the tertiary structure of the proteins. ✓

(C) The change leads to decreased protein stability because of an increased number of S-S bonds in the tertiary structure of the proteins.

(D) The change leads to increased protein stability only when the added cysteine amino acids are next to other cysteine amino acids in the primary structure.

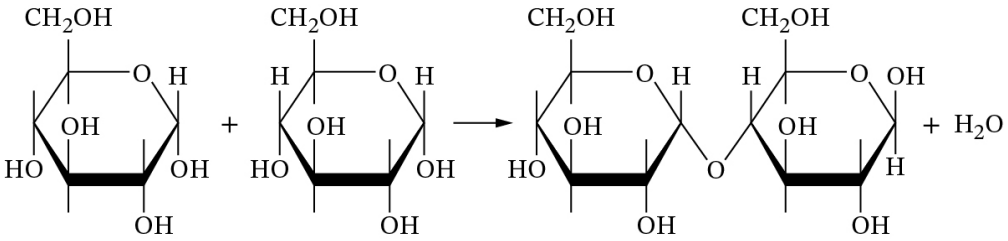
1 Chemistry of Life MCQ

Answer B

Correct. An increased number of S-S bonds are possible with the addition of more cysteine in the proteins. The S-S covalent bonds should add more structural stability to the proteins.

23. Figure 1 shows the formation of maltose (a disaccharide) from two glucose molecules.

Figure 1. Diagram of the formation of maltose



Which of the following statements best describes the molecular changes that occur during the reaction shown in Figure 1?

(A) Two hydroxyl groups are removed from each glucose molecule, and one water molecule is formed.

(B) A water molecule is added to the hydroxyl groups to form the covalent bond between the two glucose molecules.

(C) A hydroxyl group from one glucose molecule and a hydrogen atom from the other glucose molecule are removed, and one water molecule is formed. ✓

(D) A hydrogen atom in one glucose molecule and a hydroxyl group in the other glucose molecule form a hydrogen bond.

1 Chemistry of Life MCQ

24. A polypeptide is polymer of amino acids held together by peptide bonds. The process of dehydration synthesis creates these peptide bonds, as shown in Figure 1.

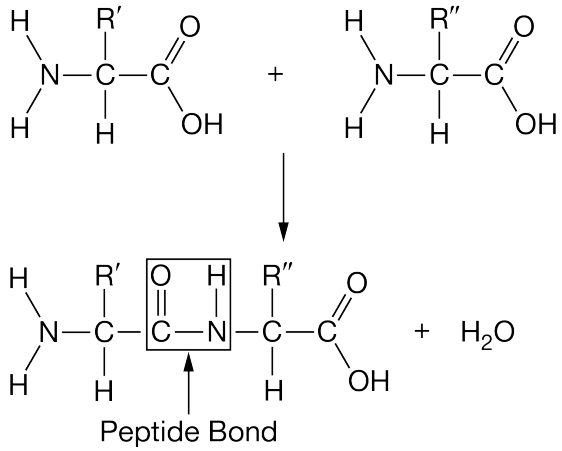


Figure 1. Amino acids are linked through the formation of peptide bonds.

As shown in Figure 1, an amino acid must have which of the following properties in order to be incorporated into a polypeptide?

(A) The ability to remain stable in the presence of water molecules

(B) An R-group that is compatible with the R-group of the last amino acid incorporated

(C) A central carbon atom that reacts with a nitrogen atom to form the peptide bond

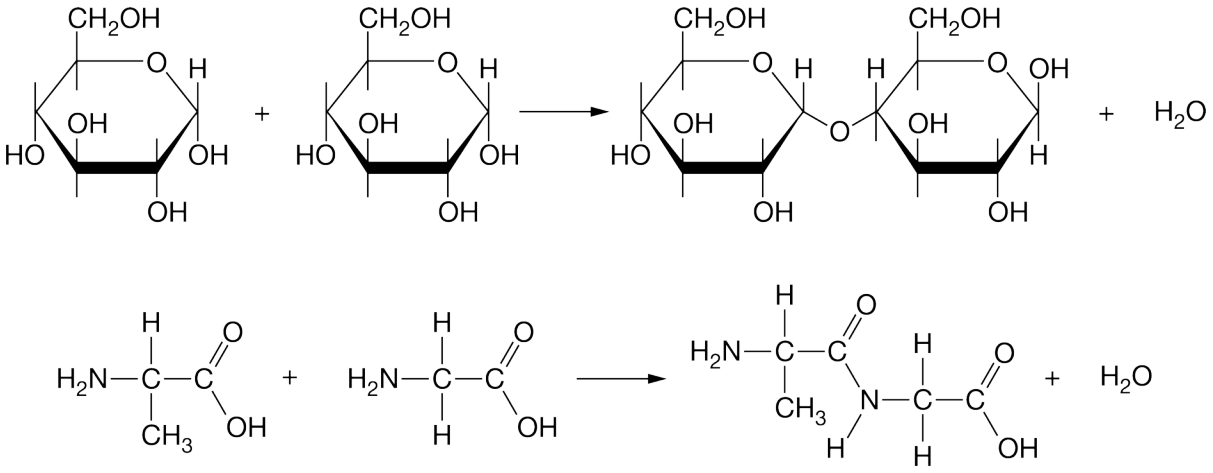
(D) The ability to form a covalent bond with both its NH_2 group and its $COOH$ group ✓

Answer D

Correct. An amino acid needs to participate in the formation of two peptide bonds in order to be part of a polypeptide. The first bond may occur when the NH_2 group interacts with the $COOH$ group of another amino acid. The second bond occurs when the $COOH$ group interacts with the NH_2 group of a third amino acid.

1 Chemistry of Life MCQ

25.



Which of the following is common feature of the illustrated reactions showing the linking of monomers to form macromolecules?

(A) Two identical monomers are joined by a covalent bond.

(B) Two different monomers are joined by a covalent bond.

(C) Monomers are joined by a covalent bond, and a water molecule is produced. ✓

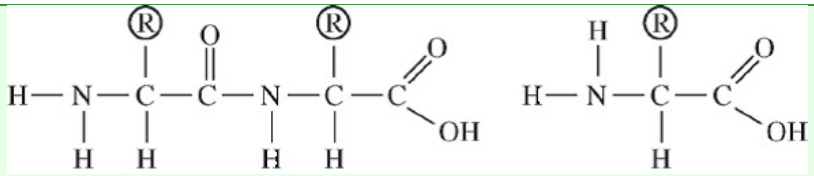
(D) Monomers are joined by ionic bonds, and a water molecule is produced.

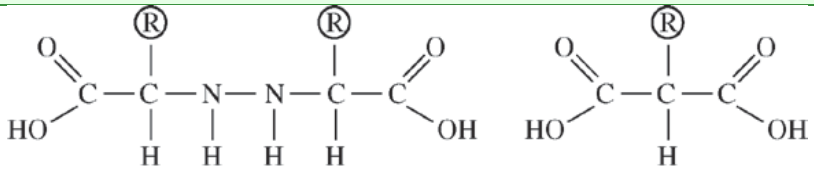
Answer C

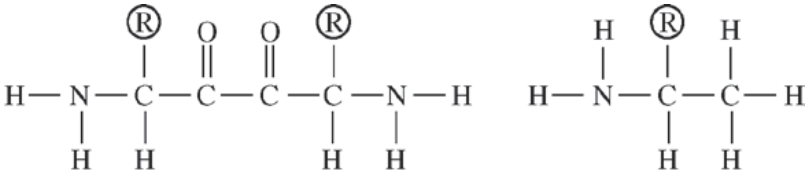
Correct. The monomers of the two reactions illustrated are joined by covalent bonds with the production of a water molecule.

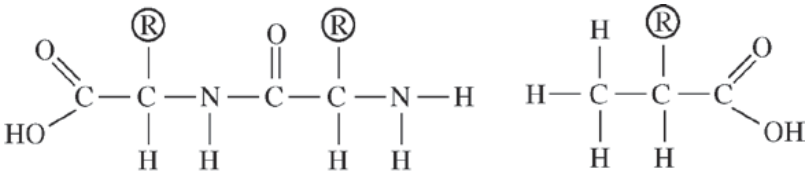
26. Which of the following correctly illustrates a dipeptide and an amino acid in the optimal position to form a tripeptide?

1 Chemistry of Life MCQ

(A) 

(B) 

(C) 

(D) 

Answer A

This option is correct. It demonstrates an ability to refine and/or interpret a representation to explain the synthesis of a biological polymer. Peptides or proteins are polymers of amino acid monomers arranged in a unique linear sequence. Each of the 20 amino acids consists of a carbon atom surrounded by an amine group (NH_2), a carboxyl group (COOH), a hydrogen, and an R (variable) group. When two amino acids are positioned so that the carboxyl group of one amino acid backbone is adjacent to the amine group of another amino acid backbone, they can join by a dehydration reaction.

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27.

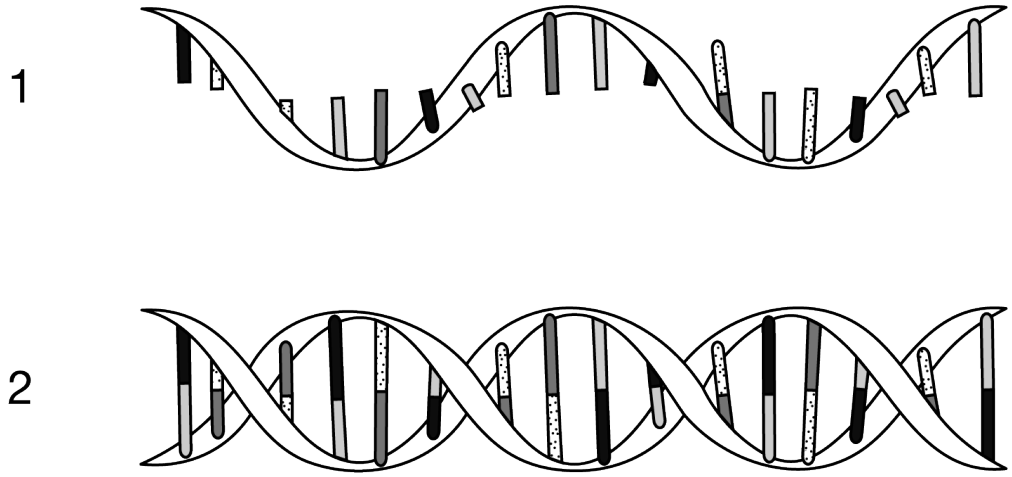


Figure 1. Molecule 1 represents RNA, and molecule 2 represents DNA.

Which of the following best describes a structural similarity between the two molecules shown in Figure 1 that is relevant to their function?

(A) Both molecules are composed of the same four nucleotides, which allows each molecule to be produced from the same pool of available nucleotides.

(B) Both molecules are composed of the same type of five-carbon sugar, which allows each molecule to act as a building block for the production of polysaccharides.

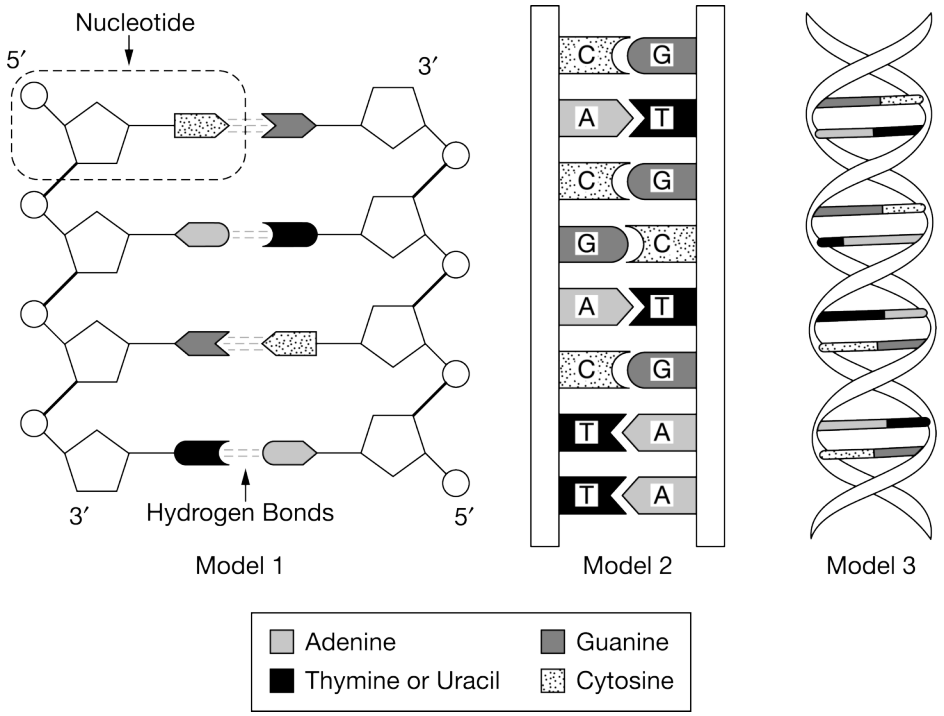
(C) Both molecules contain nucleotides that form base pairs with other nucleotides, which allows each molecule to act as a template in the synthesis of other nucleic acid molecules. ✓

(D) Both molecules contain nitrogenous bases and phosphate groups, which allows each molecule to be used as a monomer in the synthesis of proteins and lipids.

Answer C

Correct. Nucleotides form base pairs with other nucleotides. The base pairing allows a strand of RNA or DNA to act as a template in the synthesis of other nucleic acid molecules. Examples include the cellular processes of DNA replication (DNA is used as a template to make DNA), transcription (DNA is used as a template to make RNA), and reverse transcription (RNA is used as a template to make DNA).

1 Chemistry of Life MCQ



28.

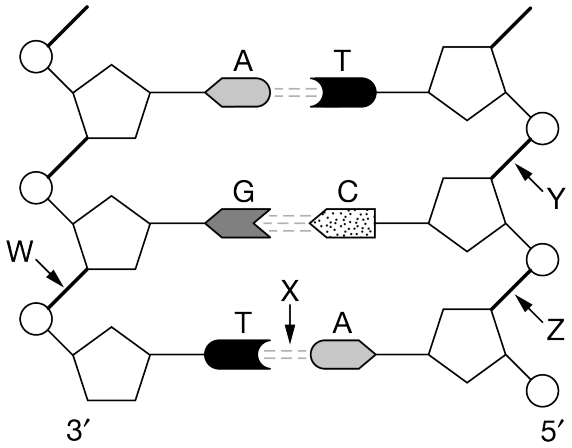


Figure 1. Four different bonds (W, X, Y, and Z) in a DNA molecule

Figure 1 represents a segment of DNA. Radiation can damage the nucleotides in a DNA molecule. To repair some types of damage, a single nucleotide can be removed from a DNA molecule and replaced with an undamaged nucleotide. Which of the four labeled bonds in Figure 1 could be broken to remove and replace the cytosine nucleotide without affecting the biological information coded in the DNA molecule?

(A) Bond X only

(B) Bond W only

(C) Bonds Y and Z at the same time ✓

(D) Bonds W and Z at the same time

1 Chemistry of Life MCQ

Answer C

Correct. By breaking bonds Y and Z at the same time, the cytosine nucleotide could be removed from the DNA molecule and replaced with an undamaged cytosine without changing the biological information stored in the DNA.

29. A student wants to modify model 1 so that it represents an RNA double helix instead of a DNA double helix. Of the following possible changes, which would be most effective in making model 1 look more like RNA than DNA ?

(A) Changing the sequence of the base pairs

(B) Changing the deoxyriboses to riboses by adding $-OH$ groups ✓

(C) Changing the shapes of the nitrogenous bases to match those shown in model 2

(D) Changing the sugar-phosphate backbone to a ribbon, as shown in model 3

Answer B

Correct. RNA contains ribose, whereas DNA contains deoxyribose. A ribose sugar has an $-OH$ group linked to the $2'$ carbon that a deoxyribose sugar does not have.

30. Which feature of model 1 best illustrates how biological information is coded in a DNA molecule?

(A) The $5'$ and $3'$ labels at the ends of each strand

(B) The labeling of the hydrogen bonds between base pairs

(C) The lines connecting sugars and phosphate groups that represent covalent bonds

(D) The linear sequence of the base pairs ✓

Answer D

Correct. The sequence of base pairs in a DNA molecule plays a central role in the coding of biological information.

1 Chemistry of Life MCQ

31. Figure 1 shows a short segment of a double-stranded nucleic acid molecule.

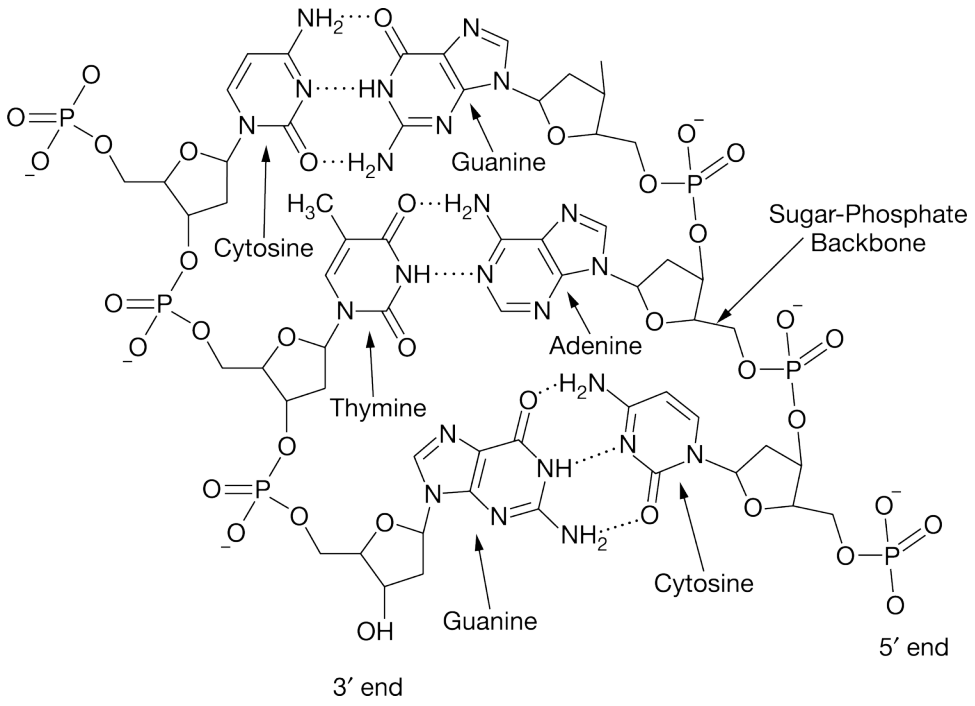


Figure 1. A short segment of a double-stranded nucleic acid molecule

Which of the following statements is correct about the molecule shown in Figure 1 ?

(A) It is RNA because of the relative direction of the two strands.

(B) It is RNA because of the number of different nucleotides found in the molecule.

(C) It is DNA because of the nature of the hydrogen bonds between guanine and cytosine.

(D) It is DNA because of the nucleotides present. ✓

Answer D

Correct. The figure indicates that one of the nucleotides is thymine. Thymine is found in DNA and not in RNA.

1 Chemistry of Life MCQ

32. Figure 1 represents a nucleic acid fragment that is made up of four nucleotides linked together in a chain.

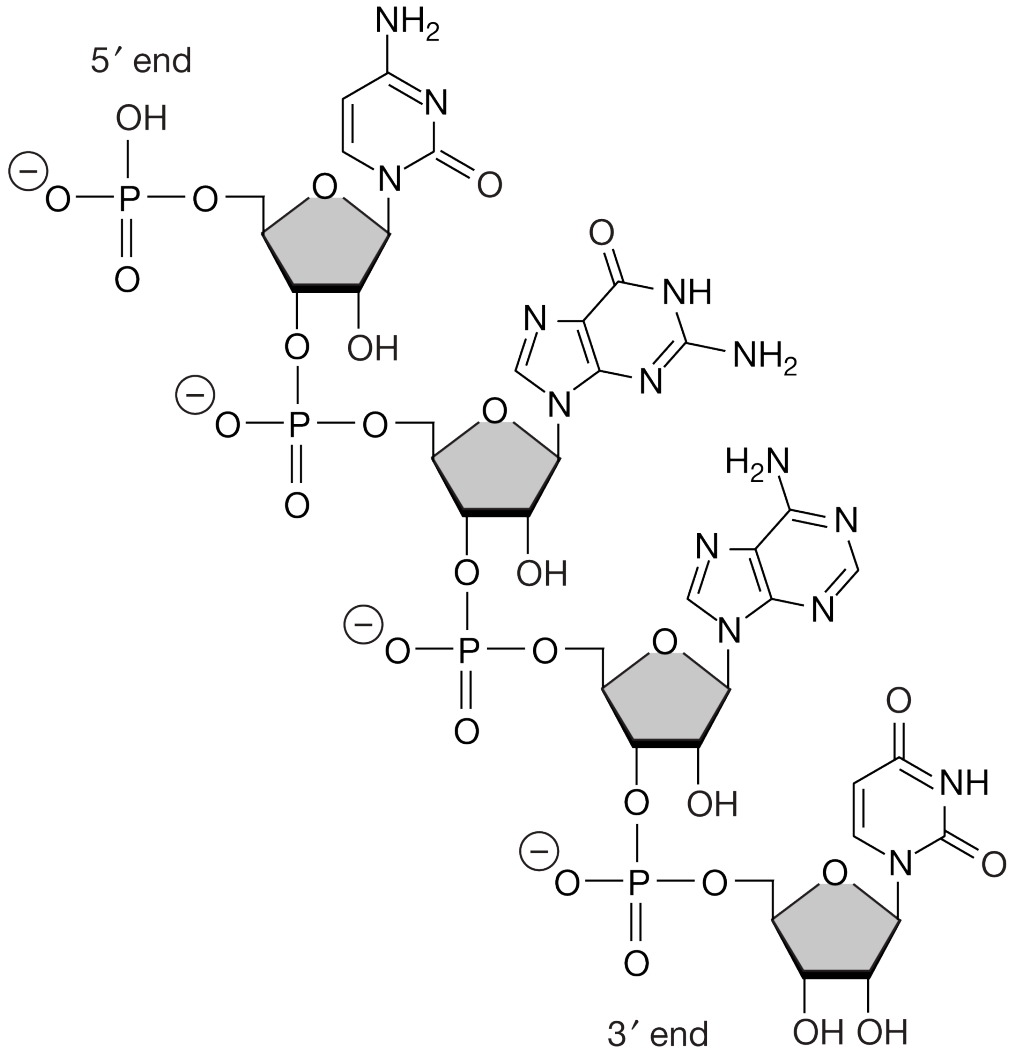


Figure 1. Nucleic acid fragment

Which of the following characteristics of Figure 1 best shows that the fragment is RNA and not DNA?

(A) The 5′ to 3′ orientation of the nucleotide chain

(B) The identity of each nitrogenous base ✓

(C) The charges on the phosphate groups

(D) The type of bond linking the nucleotides together

1 Chemistry of Life MCQ

Answer B

Correct. RNA contains uracil, whereas DNA contains thymine.

33.

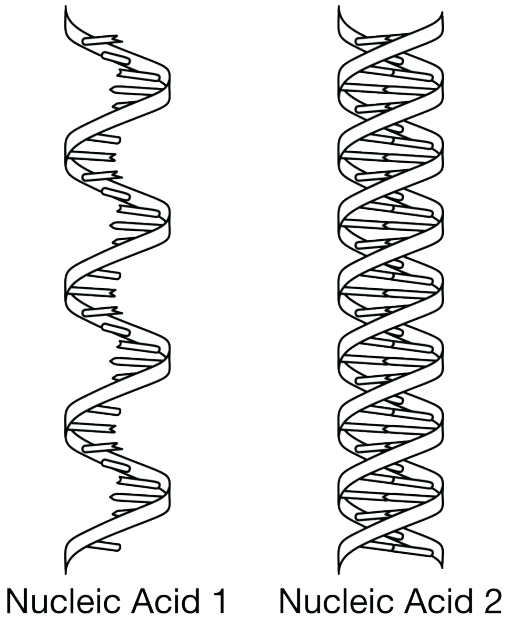


Figure 1. Nucleic acid segments

DNA and RNA are nucleic acids that can store biological information based on the sequence of their nucleotide monomers. Figure 1 shows a short segment of each of the two types of nucleic acids.

Which of the following best describes a structural difference between DNA and RNA?

(A) DNA contains four types of nitrogenous bases, whereas RNA contains only two types of nitrogenous bases.

(B) The backbone of DNA contains deoxyribose, whereas the backbone of RNA contains ribose. ✓

(C) A DNA molecule is composed of two parallel strands with the same $5'$ to $3'$ directionality, whereas an RNA molecule is composed of only one $5'$ to $3'$ strand.

(D) Phosphate groups provide rigidity to DNA, but RNA is flexible and contains no phosphate groups.

Answer B

Correct. The backbones of DNA and RNA are composed of an alternating sugar and base. In DNA, the sugar is deoxyribose, whereas the sugar in RNA is ribose.

1 Chemistry of Life MCQ



Figure 1. Domain structure of two proteins

Investigators studied the interactions between two different proteins, protein I and protein II. Each protein has two structural domains (Figure 1) that represent different functional parts of the protein. Different combinations of the protein domains were tested for their ability to bind to a known DNA sequence, interact with each other, and activate transcription. The results are shown in Table 1.

TABLE 1. INTERACTION AND FUNCTION OF COMBINATIONS OF STRUCTURAL VARIANTS OF PROTEINS I AND II

Test	Domain(s) of Protein I Used	Domain(s) of Protein II Used	DNA Binding	Protein-Protein Interaction	Transcription Activation
1	FG	LM	+	+	+
2	FG	None	+	−	−
3	None	LM	−	−	−
4	F	LM	+	−	−
5	G	LM	−	+	−
6	FG	L	+	−	−
7	FG	M	+	+	+

34. Which of the following claims is best supported by the data from test 5 in Table 1 ?

(A) Domain L is required for transcription activation.

(B) Domain F is not required for protein-protein interaction. ✓

(C) Protein-protein interaction is necessary for DNA binding.

(D) Protein-protein interaction is sufficient for transcription activation.

1 Chemistry of Life MCQ

Answer B

Correct. Based on the data for test 5 in which domain F is omitted but protein-protein interaction occurs, the G domain but not the F domain of protein I interacts with protein II. This is also confirmed in test 4 in which the F domain but not the G domain of protein I is present and in which protein-protein interaction does not occur.

35. Based on the data, which structural domain is most likely required for DNA binding?

(A) Domain F ✓
(B) Domain G
(C) Domain L
(D) Domain M

Answer A

Correct. Based on the data in Table 1, DNA binding occurs anytime both domains of protein I are present. The data for test 4 indicate that DNA binding also occurs when only domain F of protein I is present. The data for test 5 confirm that the presence of domain G is insufficient to allow DNA binding. Therefore, domain F is responsible for DNA binding by protein I.

36. Which of the following claims is best supported by the data in Table 1 ?

(A) Protein-protein interaction always results in transcription activation.
(B) DNA binding is required for protein-protein interaction.
(C) Protein-protein interaction is necessary for, but not adequate for, transcriptional activation. ✓
(D) DNA binding always results in transcription activation.

Answer C

Correct. Based on the data for tests 1 and 7, transcription activation occurs when both DNA binding and protein-protein interaction occur. The data for test 5 demonstrate that protein-protein interaction alone is insufficient for transcription activation to occur.

1 Chemistry of Life MCQ

37. Which of the following is the most likely result of testing a combination of domains G and M?

(A) No functionality

(B) DNA binding only

(C) Transcriptional activation only

(D) Protein-protein interaction only ✓

Answer D

Correct. Based on the data from tests 4 and 5, domain G (and not domain F) enables protein-protein interaction. Based on the data from tests 7 and 8, domain M (and not domain L) enables protein-protein interaction. Therefore, protein-protein interaction is expected to occur when domains G and M are combined.

---

38. A chemical binds to a protein composed of a single polypeptide chain and prevents the formation of an alpha helix that is typically formed in the absence of the chemical. Which of the following best describes the effect the chemical has on the structure of the protein?

(A) The primary structure held together by covalent bonds is affected.

(B) The secondary structure held together by hydrogen bonds is affected. ✓

(C) The secondary, tertiary, and quaternary structures are affected.

(D) All levels of protein structure are affected.

Answer B

Correct. Alpha helices are a form of secondary protein structure and are held together by hydrogen bonds. This chemical prevents alpha helices from forming, and therefore, it is most likely that the chemical affects hydrogen bonding within the local region of amino acids.

1 Chemistry of Life MCQ

39.

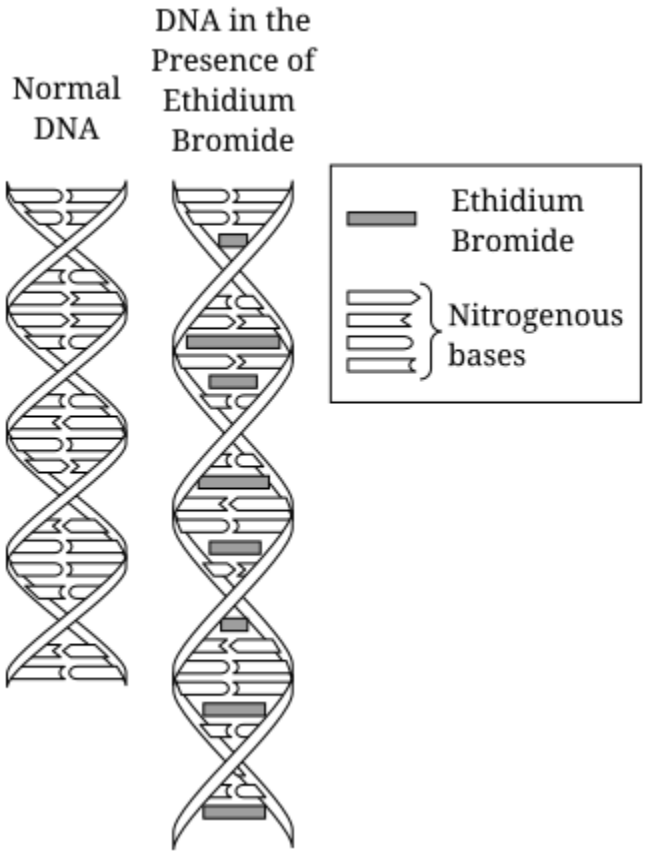


Figure 1. Model of DNA structure in the absence and presence of ethidium bromide

Ethidium bromide is a chemical that disrupts the replication of DNA. Based on the model shown in Figure 1, which of the following best describes the effect of ethidium bromide on DNA structure?

(A) It is inserted between complementary nitrogenous bases, causing the two strands of DNA to denature.

(B) It is inserted between complementary nitrogenous bases, causing hydrogen bonds to form between noncomplementary nitrogenous bases.

(C) It is inserted between sequential nitrogenous base pairs, increasing the length of the DNA molecule. ✓

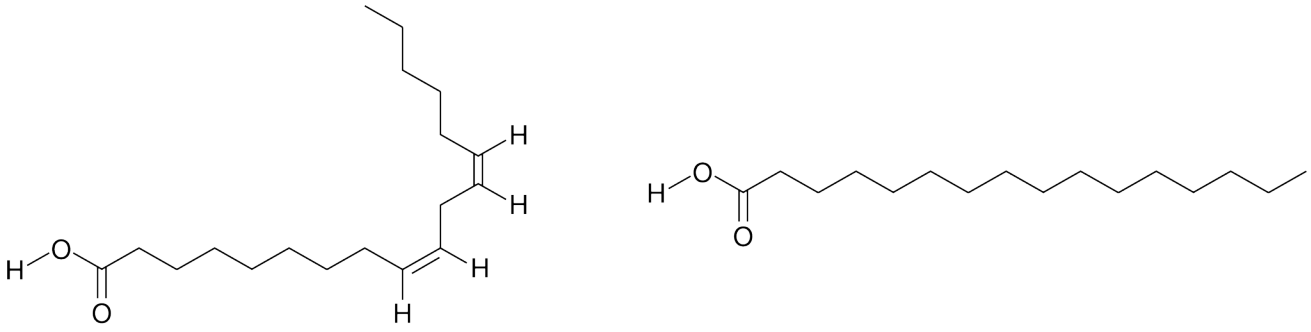
(D) It is inserted in place of certain nitrogenous base pairs, decreasing the length of the DNA molecule.

Answer C

Correct. Ethidium bromide is inserted between sequential base pairs, resulting in a longer DNA molecule.

1 Chemistry of Life MCQ

40. The molecular structures of linoleic acid and palmitic acid, two naturally occurring substances, are shown in the figure.



Linoleic Acid

Palmitic Acid

Based on the molecular structures shown in the figure, which molecule is likely to be solid at room temperature?

(A) Linoleic acid, because the absence of carbon-carbon double bonds allows the molecules to pack closely together.

(B) Linoleic acid, because the presence of carbon-carbon double bonds prevents the molecules from packing closely together.

(C) Palmitic acid, because the absence of carbon-carbon double bonds allows the molecules to pack closely together. ✓

(D) Palmitic acid, because the presence of carbon-carbon double bonds prevents the molecules from packing closely together.

Answer C

Correct. Palmitic acid is a saturated fatty acid. The absence of carbon-carbon double bonds in palmitic acid allows the molecules to pack closely together and form a solid at room temperature.

41. The fluid mosaic model of cell membranes describes proteins and steroids associated with a phospholipid bilayer. Which of the following membrane components affect membrane stability?

(A) Embedded proteins give plasma membranes less stability.

(B) Steroids, such as cholesterol, give membranes more stability. ✓

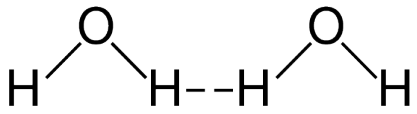
(C) Proteins in the hydrophilic region of the plasma membrane create more stability.

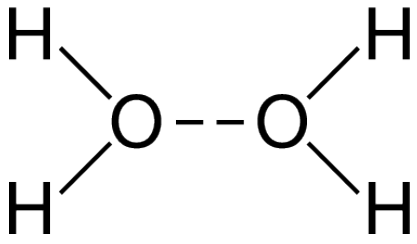
(D) Polar heads of phospholipids give plasma membranes less stability.

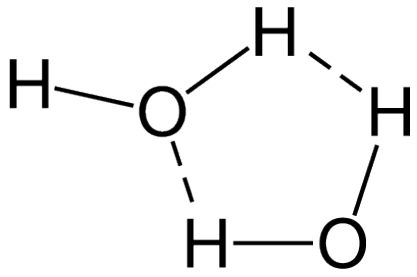
1 Chemistry of Life MCQ

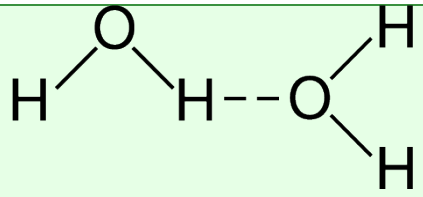
42. Water molecules are polar covalent molecules. There is a partial negative charge near the oxygen atom and partial positive charges near the hydrogen atoms due to the uneven distribution of electrons between the atoms, which results in the formation of hydrogen bonds between water molecules. The polarity of water molecules contributes to many properties of water that are important for biological processes.

Which of the following models best demonstrates the arrangement of hydrogen bonds between adjacent water molecules?

(A) 

(B) 

(C) 

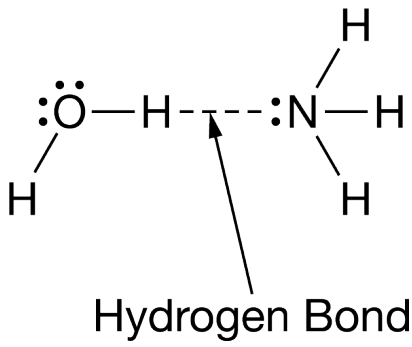
(D) 

Answer D

Correct. The hydrogen bonds between these water molecules correctly show the attractive force between the hydrogen atom of one water molecule and the oxygen atom of the adjacent water molecule.

1 Chemistry of Life MCQ

43. Water and ammonia interact to form hydrogen bonds, as represented in the figure.



Which statement best helps explain the formation of the hydrogen bond represented in the figure?

(A) The oxygen has a partial positive charge, and the nitrogen has a partial negative charge.

(B) The nitrogen has a partial negative charge, and the hydrogen attached to the oxygen has a partial positive charge. ✓

(C) The hydrogen attached to the oxygen has a partial negative charge, and the nitrogen also has a partial negative charge.

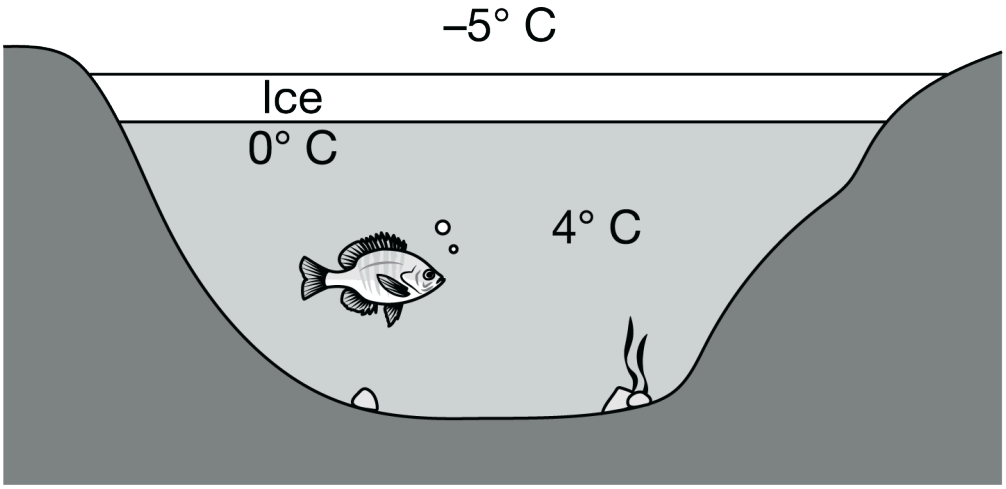
(D) The nitrogen has a partial positive charge, and the hydrogen attached to the oxygen also has a partial positive charge.

Answer B

Correct. The greater electronegativity of oxygen and nitrogen compared with hydrogen makes for unequal sharing of electrons, which results in partial negative charges associated with the oxygen and nitrogen atoms and partial positive charges associated with the hydrogen atoms in both molecules. The attraction between opposite charges results in the formation of the hydrogen bond represented in the figure.

1 Chemistry of Life MCQ

44.



As shown in the diagram, when environmental temperatures drop below freezing, a layer of ice typically forms on the surface of bodies of freshwater such as lakes and rivers.

Which of the following best describes how the structure of ice benefits the organisms that live in the water below?

(A) The water molecules in ice are closer together than those in liquid water, so the ice prevents the passage of air to the water, maintaining a constant gas mixture in the water.

(B) The water molecules in ice are closer together than those in liquid water, so the ice forms a barrier that protects the organisms in the water from the freezing air temperatures.

(C) The water molecules in ice are farther apart than those in liquid water, so the ice floats, maintaining the warmer, denser water at the lake bottom. ✓

(D) The water molecules in ice are farther apart than those in liquid water, so the ice floats, preventing the escape of gases from the liquid water.

Answer C

Correct. The water molecules in ice are farther from each other than are water molecules in liquid water, so ice is less dense than liquid water and floats on its surface, while the denser water at 4 degrees Celsius sinks to the bottom, maintaining a steady temperature all winter.

1 Chemistry of Life MCQ

45. The process of hydrogenation is used in the food industry to convert lipids that are liquids at room temperature into solids at room temperature. The process involves the chemical addition of hydrogen atoms to the fatty acid tails of lipids, breaking some or all of the carbon-carbon double bonds in the fatty acid tails.

Which of the following best explains why hydrogenation converts liquid lipids to solid lipids?

(A) Hydrogenation makes the fatty acids more saturated, enabling them to be more densely packed together. ✓

(B) Hydrogenation makes the fatty acids less saturated, enabling them to be more densely packed together.

(C) Hydrogenation makes the fatty acids more saturated, resulting in fewer lipid molecules fitting within a given space.

(D) Hydrogenation makes the fatty acids less saturated, resulting in fewer lipid molecules fitting within a given space.

Answer A

Correct. Adding hydrogen atoms results in fatty acid tails that are more saturated. The hydrocarbon chains of saturated fatty acids are straighter than those of unsaturated molecules and can therefore pack more closely together. More intermolecular forces result in higher melting points, converting the lipids to solids at room temperature.

46. One form of the genetic disorder lipodystrophy involves the lack of an enzyme that is required for the synthesis of triglycerides, a type of lipid. Triglycerides serve as an energy reserve in fat cells.

Which of the following best describes the most likely effect of lipodystrophy on cell function?

(A) There will be a decrease in phospholipids in cell membranes.

(B) There will be a decrease in cholesterol in cell membranes.

(C) There will be a decrease in fat storage in cells. ✓

(D) There will be a decrease in fatty acid synthesis in cells.

1 Chemistry of Life MCQ

47.

TABLE 1. PERCENT CHEMICAL COMPOSITION OF FOUR MACROMOLECULES

Macromolecule Sample	C (%)	O (%)	H (%)	N (%)	P (%)	S (%)
A	25	25	50	0	0	0
B	15	20	40	20	0	5
C	40	4	45	0	1	0
D	25	20	35	15	5	0

A researcher analyzed four different samples of macromolecules, where all macromolecules in each sample are of the same type. The researcher measured the percent of carbon, oxygen, hydrogen, nitrogen, phosphorus, and sulfur atoms in each sample. The results are shown in Table 1.

Which of the following claims is best supported by the data in Table 1 ?

(A) Sample A contains nucleic acids.

(B) Sample B contains protein. ✓

(C) Sample C contains nucleic acids.

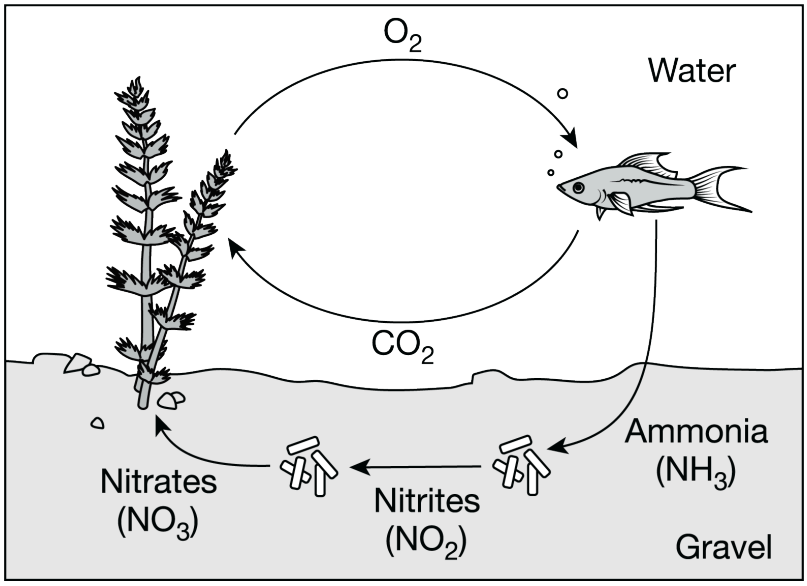
(D) Sample D contains protein.

Answer B

Correct. This sample is consistent with the atoms found in the largest quantities in proteins (C, H, O, N). A small amount of sulfur is also typically found in proteins. This sample does not have phosphorus, which would be indicative of a nucleic acid and not a protein.

1 Chemistry of Life MCQ

48. The figure shows a model of the exchange of matter between the organisms that live together in an aquarium. The model includes matter exchange between plants, fish, and bacteria. The bacteria are represented as rod-shaped organisms living in the gravel at the bottom of the aquarium.



Which of the following statements best describes how molecules released by the fish become nutrients for the plants?

(A) The carbon dioxide molecules released by the fish are converted by the bacteria to oxygen atoms, which are used by the plants to make water molecules.

(B) The oxygen molecules released by the fish are converted by the bacteria to ammonia molecules, which are used by the plants to make lipids and fatty acids.

(C) The nitrites released by the fish are converted by the bacteria to carbon dioxide molecules, which are used by the plants to make carbohydrates.

(D) The ammonia molecules released by the fish are converted by the bacteria to nitrates, which are used by the plants to make proteins and nucleic acids. ✓

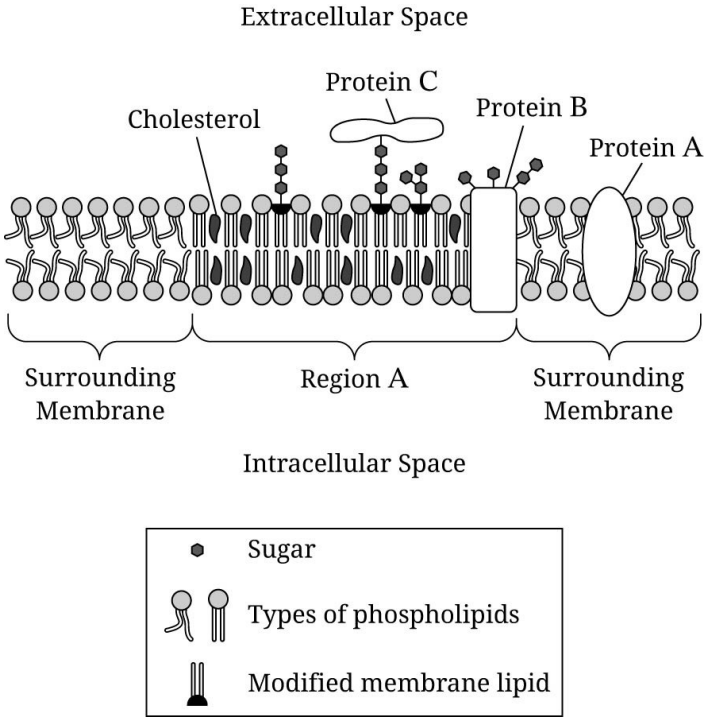
Answer D

Correct. As shown in the figure, the ammonia molecules released by the fish are converted by the bacteria to nitrites and nitrates. The plants absorb the nitrates, which are nitrogen-containing nutrients that the plants use to make the amino acids and nucleotides needed for building proteins and nucleic acids.

1 Chemistry of Life MCQ

49. Researchers have identified regions, including region A in Figure 1, of the cell membrane that have a different density and composition than the surrounding membrane.

Figure 1. Structure of a portion of the cell membrane



Which of the following best describes the chemical composition of the different types of macromolecules that make up the membrane represented in Figure 1?

(A) All the molecules contain carbon and oxygen, but only some contain nitrogen. ✓

(B) All the molecules contain carbon and hydrogen, but only some contain oxygen.

(C) All the molecules contain carbon and oxygen, but only some contain hydrogen.

(D) All the molecules contain carbon and nitrogen, but only some contain oxygen.

1 Chemistry of Life MCQ

50.

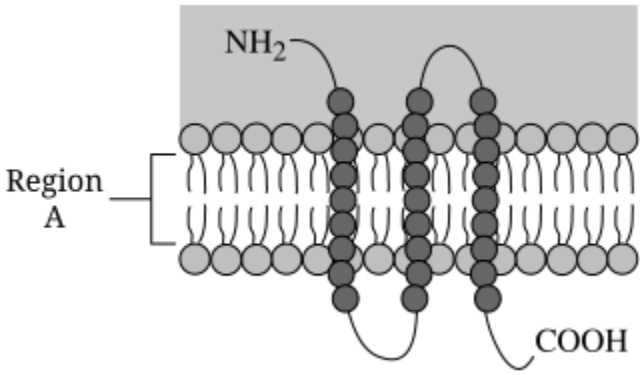


Figure 1. Structure of a membrane protein

Based on Figure 1, the amino acids in region A are most likely to have which of the following characteristics?

(A) Most amino acids will be hydrophobic because they interact most favorably with the phospholipid tails. ✓

(B) Most amino acids will be hydrophilic because they interact most favorably with the phospholipid heads.

(C) Most amino acids will be ionic amino acids because they interact most favorably with the phospholipid tails.

(D) Most amino acids will be polar amino acids because they interact most favorably with the phospholipid heads.

Answer A

Correct. Region A brackets the hydrophobic tails of the membrane phospholipids. In order for a protein to be embedded in the membrane among the phospholipid tails, the amino acids of the portion of the protein in region A must also be hydrophobic.

51. Which of the following statements best describes the structure of complex carbohydrates?

(A) Several polysaccharide chains are linked via hydrogen bonds.

(B) Monosaccharides are connected by covalent bonds. ✓

(C) Several polysaccharide chains are linked via ionic bonds.

(D) Monosaccharides are joined by peptide bonds.

1 Chemistry of Life MCQ

52. A small protein is composed of 110 amino acids linked together in a chain. As shown in Figure 1, the first and last five amino acids in the chain are hydrophobic (have nonpolar and uncharged R-groups), whereas the remaining 100 amino acids are hydrophilic (have charged or polar R-groups). The nature of the R-group determines if the amino acid is hydrophobic or hydrophilic.

A mutation results in the production of a version of the small protein that is only 105 amino acids long, as shown in Figure 2. Five of the hydrophobic amino acids are missing from one end of the chain.

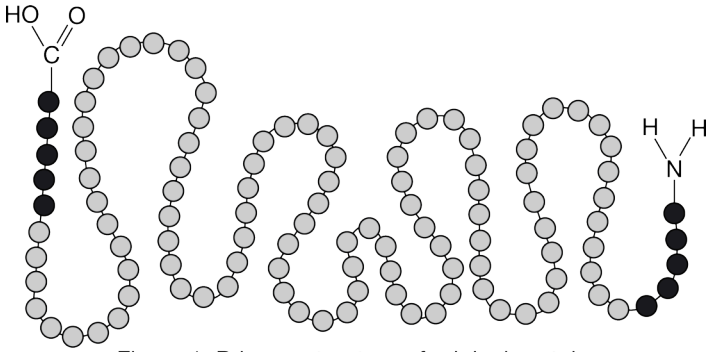


Figure 1. Primary structure of original protein

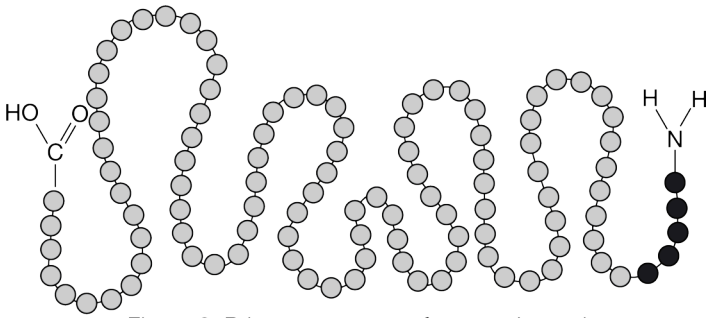


Figure 2. Primary structure of mutated protein

Hydrophilic Amino Acid
Hydrophobic Amino Acid

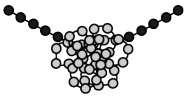
Which of the following best depicts the tertiary structures of the two proteins in water? The diagrams in the options are not drawn to the same scale as those in Figure 1 and Figure 2.

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(A)

Original Protein

Mutated Protein

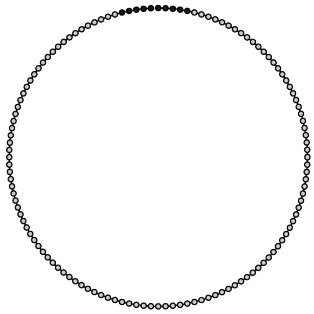
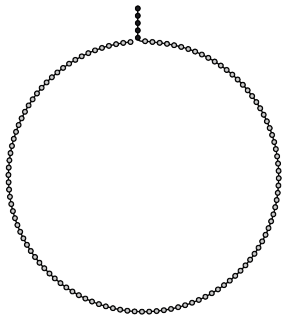
(B)

Original Protein

Mutated Protein

(C)

Original Protein

Mutated Protein

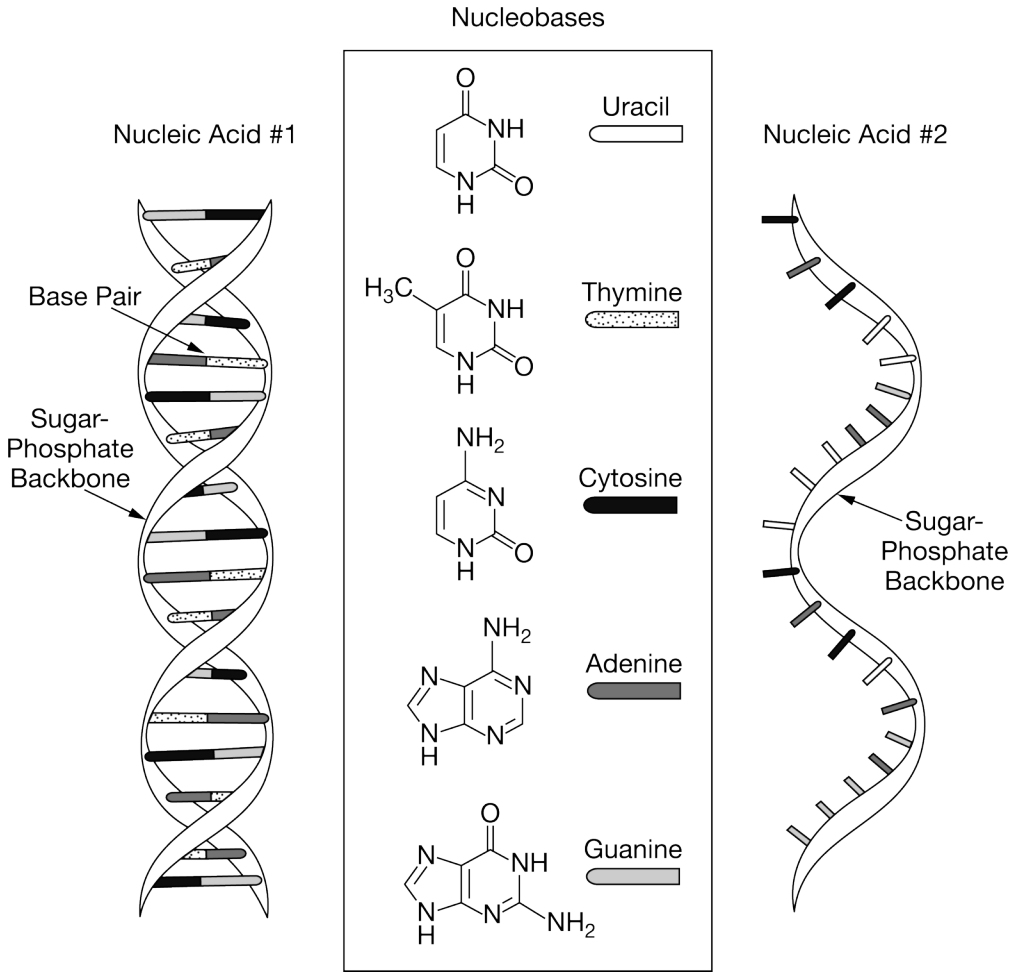
(D)

Original Protein

Mutated Protein

Answer A

Correct. The hydrophilic middle of the protein chain would be attracted to the water molecules, and the hydrophobic ends would be repelled. Hydrophobic portions of proteins in water tend to be found near the center of the tertiary structure, as far from the water as possible.

1 Chemistry of Life MCQ

53.



Which of the following conclusions is most clearly supported by the representations of nucleic acid #1 and nucleic acid #2 ?

(A) Nucleic acid #1 contains only purines, whereas nucleic acid #2 contains only pyrimidines.

(B) Nucleic acid #1 contains the sugar ribose, whereas nucleic acid #2 contains the sugar deoxyribose.

(C) Nucleic acid #1 contains positively charged phosphate groups, whereas nucleic acid #2 does not.

(D) Nucleic acid #1 contains adenine-thymine base pairs, whereas nucleic acid #2 does not. ✓

Answer D

Correct. The representation of nucleic acid #1 shows adenine-thymine base pairs, whereas the representation of nucleic acid #2 shows only uracil in the nucleotide chain and does not show any base pairing.

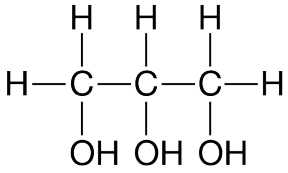
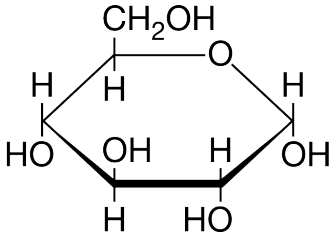
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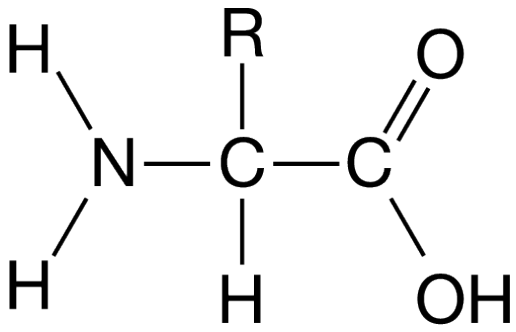
54. Students conducted a controlled experiment to investigate whether sawdust provides enough nutrients to support plant growth. The students separated ten nearly identical sunflower seedlings into two groups. They grew the seedlings in the first group in potting soil and the seedlings in the second group in sawdust composed mostly of cellulose. After twenty days, the students recorded observations about the seedlings in each group. The students' observations are presented in the table.

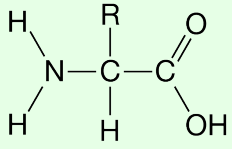
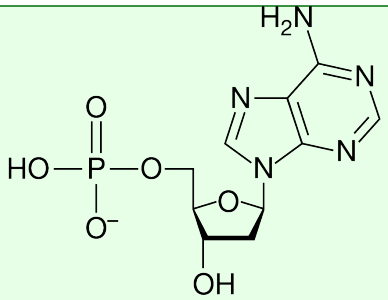
Treatment Group	Mean Seedling Height	Observations
Seedlings in potting soil	18.5 cm	The leaves have a dark green color and are normal in size.
Seedlings in sawdust	4.8 cm	The leaves have a grayish color and are small in size.

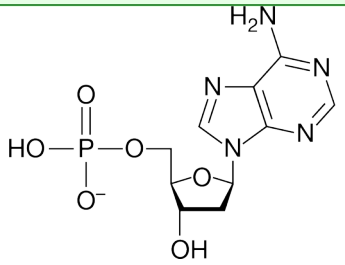
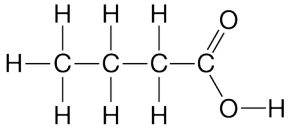
The observed differences between the groups most likely resulted from differences in the ability of the seedlings to produce which of the following monomers?

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(A)  and 

(B)  only

(C)  and 

(D)  and 

Answer C

Correct. Sawdust will not provide sufficient amounts of nitrogen and phosphorus, resulting in the seedlings having a reduced ability to produce both of the monomers shown (amino acids and nucleotides).

1 Chemistry of Life MCQ

55. Figure 1 represents a common process that occurs in organisms.

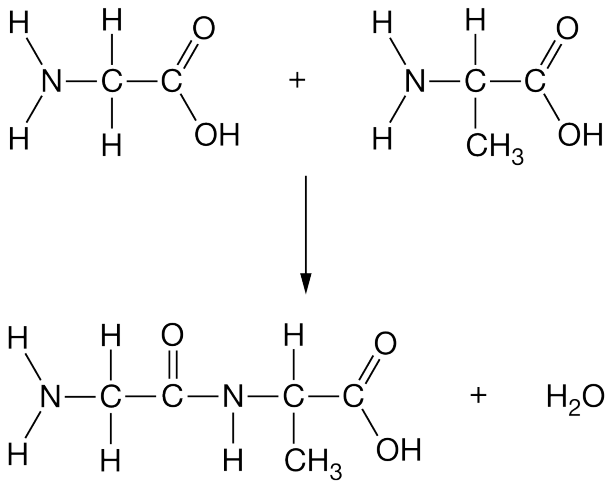


Figure 1. Structural formula for a common biological reaction

Which of the following is an accurate description of the process shown in Figure 1 ?

(A) The linking of amino acids with an ionic bond as an initial step in the protein synthesis process

(B) The formation of a more complex carbohydrate with the covalent bonding of two simple sugars

(C) The hydrolysis of amino acids with the breaking of covalent bonds with the release of water

(D) The formation of a covalent peptide bond in a dehydration synthesis reaction ✓

Answer D

Correct. This reaction combines two smaller amino acids to form a dipeptide (a step in the formation of a protein) with the release of water. The presence of nitrogen is consistent with both amino acids and proteins. The combining of smaller molecules to form a larger molecule with the release of water is a dehydration synthesis reaction.

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56.

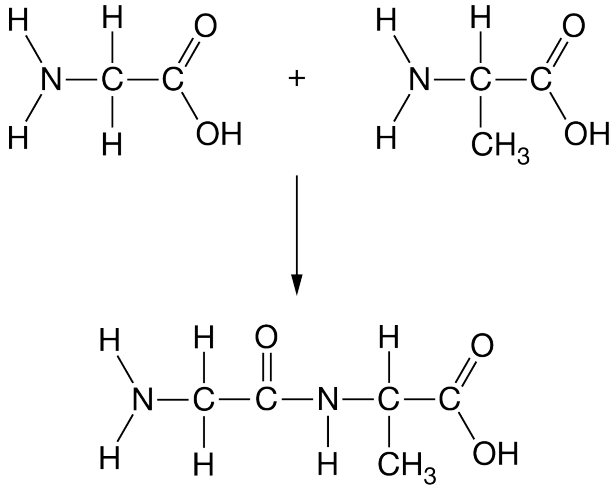


Figure 1. Formation of a peptide bond

Which of the following best describes the formation of the bond shown in Figure 1 ?

(A) An ionic bond is formed between a carbon atom of one amino acid and the nitrogen atom of the other amino acid.

(B) An ionic bond is formed when the negative charge of an OH group is balanced by the positive charge of a hydrogen ion.

(C) A covalent bond is formed between a carbon atom and a nitrogen atom along with the formation of H_2O . ✓

(D) A covalent bond is formed that replaces the hydrogen bond between the OH group and the H atom.

Answer C

Correct. Even though the water molecule that is produced is not shown in Figure 1, it shows the formation of the peptide bond and the missing H and OH.

57. Which of the following best describes the role of phospholipids?

(A) Phospholipids provide structural stability to animal cell membranes.

(B) Phospholipids support physiological functions, including growth.

(C) Phospholipids provide energy storage and support cell function.

(D) Phospholipids form the lipid bilayer found in cell membranes. ✓

1 Chemistry of Life MCQ

58. Phosphorous (P) is an important nutrient for plant growth. Figure 1 shows *Arabidopsis thaliana* plants grown under phosphorus-sufficient (left) and phosphorus-starved (right) conditions for six weeks.

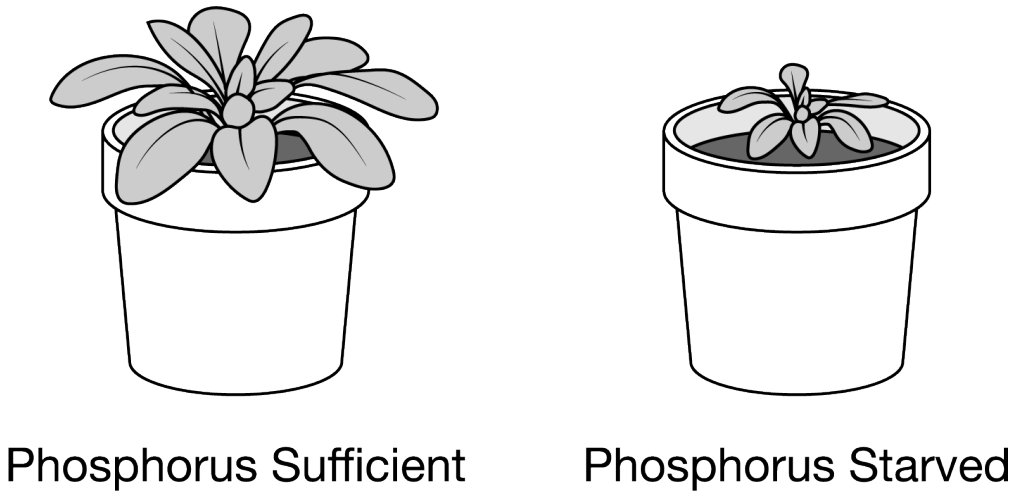


Figure 1. *Arabidopsis thaliana* plants grown for six weeks.

Which of the following is the most likely reason for the difference in leaf growth?

(A) The phosphorus-starved plant was unable to synthesize both the required proteins and lipids, limiting growth.

(B) The phosphorus-starved plant was unable to synthesize both the required proteins and carbohydrates, limiting growth.

(C) The phosphorus-starved plant was unable to synthesize both the required nucleic acids and lipids, limiting growth. ✓

(D) The phosphorus-starved plant was unable to synthesize both the required carbohydrates and nucleic acids, limiting growth.

Answer C

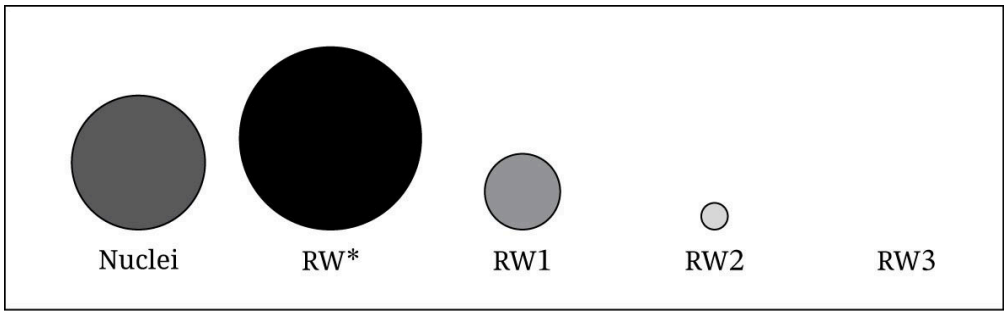
Correct. Phosphorus is used to make nucleic acids and certain lipids. Without phosphorus atoms, nucleic acids and lipids cannot be made for the plant to use for growth.

1 Chemistry of Life MCQ

59. A researcher conducted an experiment in a brightly lit aquarium containing a large population of single-celled green algae. Water containing radioactive phosphorus was added to the aquarium. After 7 days, the researcher took a water sample containing the algae and separated the algae from the radioactive water (RW*).

The researcher added deionized water to rinse the algae and saved the rinse water and then repeated that process twice. The rinse water from each of the three rinses was saved as RW1, RW2, and RW3, respectively. The researcher also removed and kept the nuclei of the algal cells. The nuclei were mixed with deionized water, and a drop of the mixture was placed on a piece of filter paper. A drop of the radioactive water and a drop of each of the rinse water samples were also placed on the same paper. These samples were exposed to X-ray to determine their radioactivity. The results after 24 hours are shown in Figure 1.

Figure 1. Radioactivity of saved algal water samples. Darker color and larger circle size indicate a higher level of radioactivity.



Based on Figure 1, which of the following statements best explains why the nuclei sample had a higher level of radioactivity than any of the rinse water samples?

(A) Phosphorus is used in the building of nucleic acids in the nucleus. ✓

(B) The nuclei sample was contaminated with radioactive water.

(C) Radioactive phosphorus is dissolved from rocks and becomes part of the sugar molecules created during photosynthesis.

(D) Radioactive phosphorus is dissolved from rocks and is used to create cholesterol stored in the nucleus.

60. Which of the following statements best describes the formation of complex carbohydrates?

(A) Monosaccharides are converted to simple sugars.

(B) Monomers of polysaccharides are linked together through peptide bonds.

(C) Nitrogen-containing bases are used to build complex carbohydrates.

(D) Monosaccharides are joined together through covalent bonds. ✓

61. Which of the following best describes the structures of carbohydrates?

1 Chemistry of Life MCQ

(A) They only occur as disaccharides.

(B) They occur as monomers, chains of monomers, and branched structures. ✓

(C) They only occur as long and branched structures.

(D) They occur as chains of monomers that hydrogen bond with complementary chains of monomers.

Answer B

Correct. Carbohydrates occur in many forms. Glucose is an example of a monomer, sucrose is a chain of two carbohydrate monomers, and starch is an example of a carbohydrate with a branched structure.

62. The synthesis of protein or carbohydrate polymers always produces which of the following as a byproduct?

(A) ATP

(B) Oxygen

(C) Carbon dioxide

(D) Urea

(E) Water ✓

63. Which of the following statements best describes a difference between proteins and nucleic acids?

(A) Proteins contain sulfur, while nucleic acids do not. ✓

(B) Proteins contain phosphorus, while nucleic acids contain sulfur.

(C) Both proteins and nucleic acids contain sulfur as a key element.

(D) Both proteins and nucleic acids contain nitrogen and sulfur.

1 Chemistry of Life MCQ

64. Polypeptides are continuously being formed and degraded. One of these processes is shown.

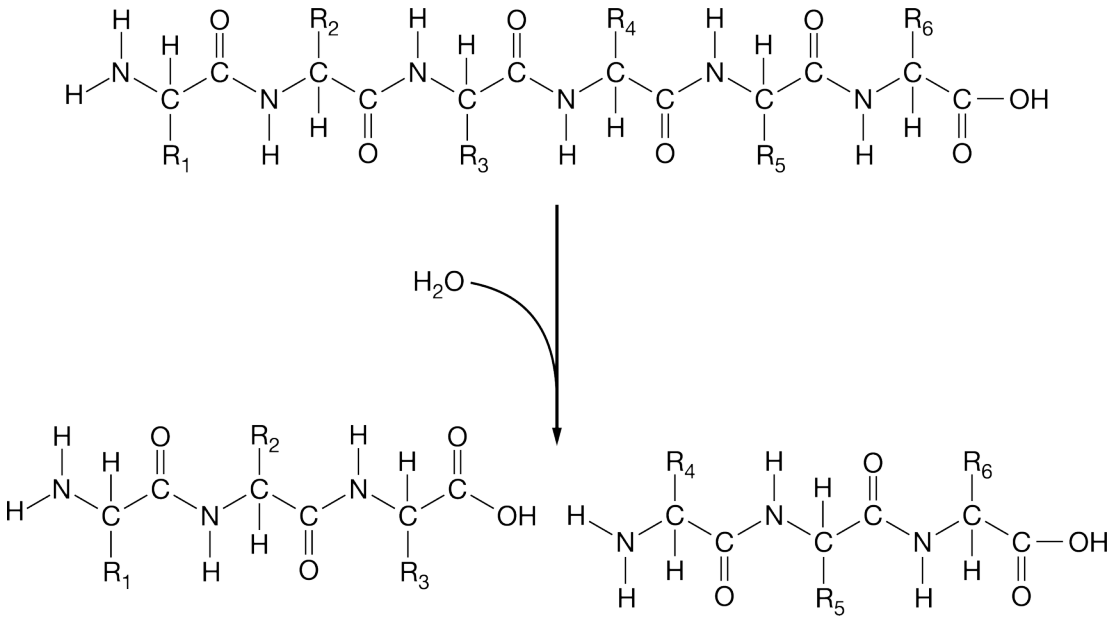


Figure 1. Polypeptide reaction

Which statement is the most accurate description of the reaction shown in Figure 1?

(A) It represents monomers linked by dehydration synthesis.

(B) It represents a polypeptide chain that folds to form the tertiary structure.

(C) It represents a polypeptide chain that is denatured into the primary structure.

(D) It represents a polypeptide chain that is broken down through a hydrolysis reaction. ✓

Answer D

Correct. The reaction shows the polypeptide being split into two smaller units with the addition of a hydroxide ion and a hydrogen ion from water. This facilitates the breaking of the bond holding the polypeptide together.

1 Chemistry of Life MCQ

For following group of questions first study the description of the situation or data and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

In the first step of an experiment, rat liver cells were exposed for 5 minutes to amino acids labeled with a radioactive isotope. The cells were then washed to stop any further incorporation of radioactive amino acids. The cells were sampled periodically thereafter, and the radioactivity of a certain protein (protein X) was measured in various cell components, as shown below.

RADIOACTIVE COUNTS (PROTEIN X)

Sampling Time (minutes)	Cell Component						
	Endoplasmic Reticulum	Golgi Apparatus	Cytoplasm	Nucleus	Mitochondria	Lysosomes	Extracellular Space
5	980	20	50	4	2	6	2
20	135	760	20	3	5	120	9
45	50	195	4	6	2	800	3
120	15	30	8	6	8	975	4

65. What is the most likely identity of protein X ?

(A) An enzyme that is involved in transcription

(B) An electron transport protein that participates in respiration reactions

(C) A protein secreted from liver cells into the blood stream that permits proper capillary function

(D) An enzyme that participates in the degradation and recycling of cell components ✓

(E) A building block of the microtubules that are part of the cytoskeleton

66. A researcher is studying a specific type of cellular defect in animal cells. She notices that cells with this defect have reduced energy storage and decreased cell membrane stability.

Which of the following predicts the type of cellular defect that is the most likely cause of the researcher's observation?

(A) A defect in fatty acid synthesis, reducing the formation of fats and phospholipids ✓

(B) A defect in cholesterol synthesis, leading to reduced cholesterol in cell membranes

(C) A defect in steroid hormone production, leading to a depletion of energy reserves

(D) A defect in unsaturated fatty acid production, affecting membrane fluidity

67. Which of the following best describes a characteristic of saturated fatty acids?

1 Chemistry of Life MCQ

(A) They contain at least one double bond between carbon atoms.

(B) They have a kinked carbon chain structure.

(C) They contain only single bonds between carbon atoms. ✓

(D) They are more liquid at room temperature than unsaturated fatty acids

Answer C

Correct. Saturated fatty acids contain only single bonds between carbon atoms. This characteristic allows the carbon chains to pack tightly together, giving saturated fats their solid nature at room temperature.

68.

Segment 1:	5′ - ATATGAGTAGT - 3′ 3′ - TATACTCATCA - 5′	Segment 2:	5′ - GCGCAGACGAC - 3′ 3′ - CGCGTCTGCTG - 5′

The sequences for two short fragments of DNA are shown above. Which of the following is one way in which these two segments would differ?

(A) Segment 1 would not code for mRNA because both strands have T, a base not found in RNA.

(B) Segment 1 would be more soluble in water than segment 2 because it has more phosphate groups.

(C) Segment 1 would become denatured at a lower temperature than would segment 2 because A-T base pairs have two hydrogen bonds whereas G-C base pairs have three. ✓

(D) Segment 1 must be from a prokaryote because it has predominantly A-T base pairs.

Answer C

This option is correct. It demonstrates an understanding of the relationship between structure and function at the molecular level and the ability to make predictions about how change(s) in structure affect functionality. Because G-C base pairs have three hydrogen bonds, they are more stable structurally. Thus, because DNA Segment 2 has more G-C base pairs than DNA Segment 1, Segment 2 is more stable and thus would denature at higher temperatures than Segment 1.

69. Which of the following best explains why surface tension is created on the surface of bodies of water?

1 Chemistry of Life MCQ

(A) Water molecules are attracted to each other because the hydrogen atoms donate electrons to oxygen atoms.

(B) Water molecules are attracted to each other because the sharing of electrons between hydrogen and oxygen is unequal. ✓

(C) Water molecules are attracted to substances dissolved in the water because most dissolved substances are positively charged.

(D) Water molecules are attracted to substances dissolved in the water because most dissolved substances are negatively charged.

Answer B

Correct. The unequal sharing of electrons between hydrogen and oxygen in a water molecule results in a partial positive charge and a partial negative charge on the hydrogen and oxygen atoms, respectively. These partial charges attract other water molecules, resulting in hydrogen bond formation and cohesive forces between the water molecules, especially at the surface of the water.

70. Figure 1 is a diagram of water molecules at the air-water interface at the surface of a pond.

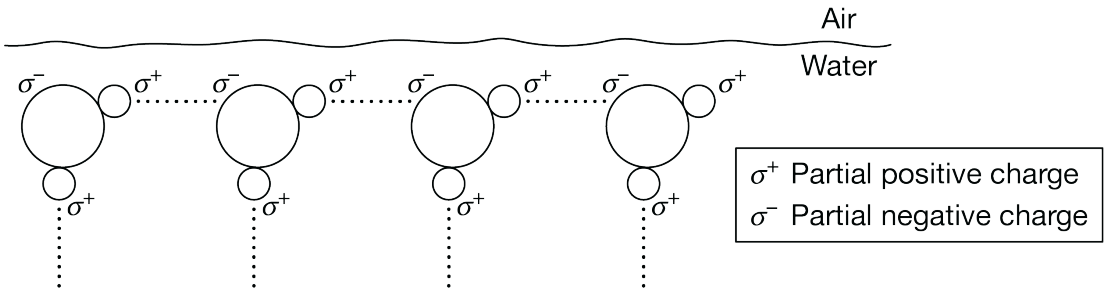


Figure 1. Alignment of water molecules at air-water interface

Based on Figure 1, which of the following best describes how the properties of water at an air-water interface enable an insect to walk on the water's surface?

(A) Covalent bonds between water molecules and the air above provide cohesion, which causes tiny bubbles to form under the feet of the insect.

(B) Ionic bonds between molecules at the surface of the water provide an electric charge, which attracts the feet of the insect, keeping it on the surface.

(C) Polar covalent bonds between molecules at the surface of the water provide adhesion, which supports the weight of the insect.

(D) Hydrogen bonds between molecules at the surface of the water provide surface tension, which allows the water surface to deform but not break under the insect. ✓

1 Chemistry of Life MCQ

Answer D

Correct. Hydrogen bonds form between water molecules due to the partial charges produced as a result of the polarity of the water molecule. The strength of the attraction between the water molecules allows organisms to walk across the surface without breaking the hydrogen bonds.

71. Which of the following best describes how amino acids affect the tertiary structure of a protein?

(A) The number of amino acids determines the tertiary structure of the protein.

(B) The interactions of the different R-groups with other R-groups and with their environment determine the tertiary structure of the protein. ✓

(C) The R-group of the last amino acid that is added to a growing polypeptide chain determines the next amino acid that is added to the chain.

(D) The sequence of the amino acids in the polypeptide chain determines the protein's primary structure but has no effect on its tertiary structure.

Answer B

Correct. The tertiary structure is determined by the attractions, repulsions, and bonding between different amino acids. Changing the sequence of even two amino acids may affect the tertiary shape of the protein.

72. Researchers analyzed the composition of cell membranes of bacteria living in different environments. Thermophiles are bacteria that live at extremely high temperatures, typically at or above $70^{\circ}C$.

Which of the following best predicts the effect of a greater proportion of saturated fatty acids, as compared with unsaturated fatty acids, on the stability of the cell membrane in thermophiles?

(A) A greater proportion of saturated fatty acids has no effect on cell membrane stability because both saturated and unsaturated fatty acids are hydrophobic.

(B) A greater proportion of saturated fatty acids leads to increased membrane stability at lower temperatures because saturated fatty acids increase the stability of the membrane.

(C) A greater proportion of saturated fatty acids leads to decreased membrane stability because saturated fatty acids decrease the fluidity of the membrane.

(D) A greater proportion of saturated fatty acids leads to increased membrane stability at higher temperatures because saturated fatty acids pack tightly together. ✓

Answer D

Correct. Saturated fatty acids pack tightly, which reduces membrane fluidity. This is critical for maintaining membrane stability in thermophiles exposed to extremely high temperatures.

1 Chemistry of Life MCQ

73. In vascular plants, water flows from root to leaf via specialized cells called xylem. Xylem cells are hollow cells stacked together like a straw. A student explains that evaporation of water from the leaf pulls water up from the roots through the xylem, as shown in Figure 1.

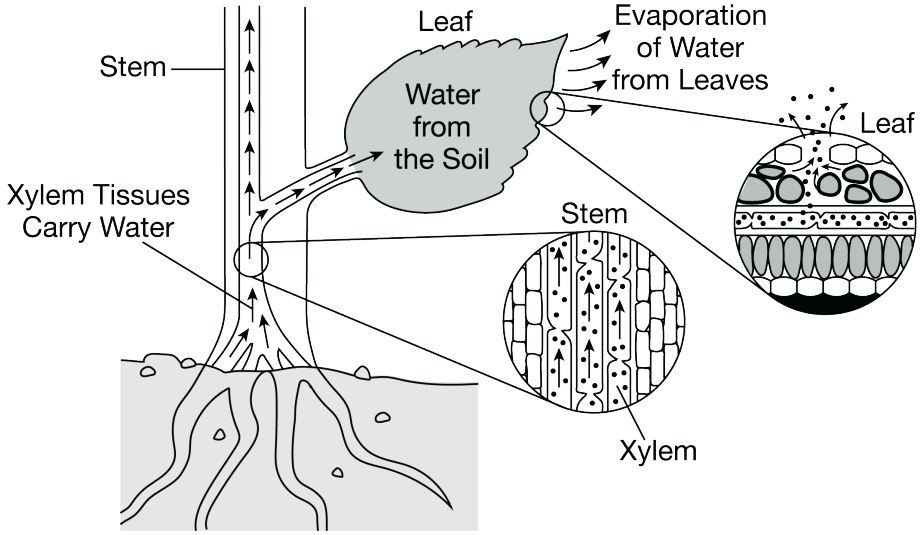


Figure 1. Model of water movement through the xylem, with magnified models of water movement in the stem and leaf.

Which statement describes how water is pulled up through the xylem to the leaves of the plant?

(A) As water exits the leaf, hydrogen bonding between water molecules pulls more water up from below. ✓

(B) As water exits the leaf, signals are sent to the roots to pump more water up to the leaves through the xylem by adhesion.

(C) Evaporation from the leaf decreases the hydrogen bonds that form between the water molecules in the xylem, which helps the water molecules to be pulled up the xylem.

(D) Evaporation of water from the leaf increases the hydrogen bonds that form between water molecules in the air, providing the energy for transport.

Answer A

Correct. As water exits the leaf, hydrogen bonding pulls more water molecules up through the leaf and xylem by cohesion.

74. Which of the following is most directly responsible for water's unique properties?

1 Chemistry of Life MCQ

(A) It contains oxygen atoms.

(B) It contains hydrogen atoms.

(C) It is an ionic compound.

(D) It forms hydrogen bonds. ✓

(E) It is nonpolar.